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Large-scale deep learning based binary and semantic change detection in ultra high resolution remote sensing imagery: From benchmark datasets to urban application

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ABSTRACT

With the acceleration of urban expansion, urban change detection (UCD), as a significant and effective approach, can provide the change information with respect to geospatial objects for dynamic urban analysis. In recent years, through the use of machine learning and artificial intelligence, change detection methods have gradually developed from the traditional pixel-based comparison methods in the 1980s to data-driven deep learning methods. Deep learning methods have huge advantages in the application of remote sensing big data, by virtue of their huge feature extraction and expression capabilities. Many change detection datasets have been released to meet the requirements of deep learning. However, the existing datasets suffer from three bottlenecks: (1) the volume of the datasets is small, which can easily cause overfitting; (2) most datasets have a spatial resolution of meters, making it difficult to detect changes in small objects because there are multi-scale objects in urban areas; and (3) most of the datasets have been designed for binary change detection (BCD), and lack semantic annotation, so that they cannot be used to obtain the direction of the change or to analyze the type of change, for further application in urban areas. Therefore, it is difficult to apply these datasets to detect large-scale urban semantic changes in complex environments. To address these issues, a large-scale ultra high resolution ($0.1\,\mathrm{m}$) UCD dataset for deep learning based BCD and semantic change detection (SCD) is introduced in this article, which is named the Hi-UCD dataset. We selected an area of $102\,\mathrm{m}^2$ in Tallinn, the capital of Estonia, as the study area. There are a total of 40800 pairs of 512 × 512 patches, nine types of land cover and 48 types of semantic change in the Hi-UCD dataset. We developed three metrics–binary consistency, change area consistency and no-change area consistency–to evaluate the semantic consistency of the SCD methods from different aspects. A comprehensive analysis and investigation is provided in this article after we benchmarked this dataset using deep learning methods for BCD and SCD. We also found that the unchanged samples can help distinguish the changed area, and HRNet used as the backbone to construct a multi-task model can perform well in the Hi-UCD dataset. Meanwhile, the visualization results obtained with the Hi-UCD test set, which is a large geographic area covering $54\,\mathrm{km}^2$, are shown to reflect the real-world urban application scenarios. The experimental results show that the Hi-UCD dataset is a challenging yet useful benchmark dataset, which can be used for analyzing large-scale refined urban changes.

1. Introduction

The purpose of change detection is to discover land-cover changes and interpret the process of feature change, where the results determine which objects have changed, both when and where. This information can provide help for urban construction (Huang et al., 2017; Daudt et al., 2018b; Amini Amirkolaee and Arefi, 2019; Tian et al., 2021), disaster emergency response (Vetrivel et al., 2018; Qin et al., 2018; Liu et al., 2019a; Zhang et al., 2020b), environmental protection (Bindschadler et al., 2010; Tian et al., 2014; Zhu et al., 2019; Ochtyra et al., 2020), and other applications (Jin et al., 2017; Hu et al., 2018; Maiya and Babu, 2018; Liu et al., 2019b), thereby assisting with decision-making. As the process of urbanization has accelerated in recent years, how to better detect urban changes has become a research focus. Urban

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change is closely related to human activities. Carrying out urban change detection (UCD) can help us to understand a series of problems in urban development, including the urban heat island effect, environmental pollution, and economic development.

Due to the complexity of urban scenes and the various types of objects, urban changes have unique characteristics, which bring challenges to UCD. In Fig. 1, there are four typical ground objects and some examples of their changes in the urban area, which show that urban changes have the following characteristics:

1. *Complex and diverse types of urban changes.* From Fig. 1(a), we can see that there are various categories of urban objects, including natural objects (e.g., vegetation and water) and artificial objects (e.g., roads and buildings), with strong regional heterogeneity and inter-class heterogeneity. Even objects of the same class, such as roads with sidewalks, highways, and urban greenways, may have different geometric shapes, spectral features, and local features. Highways are wide and straight, while urban greenways are narrow and winding. The complicated composition and distribution of the ground objects results in diverse types of changes. If there are n kinds of ground objects, theoretically, $n \times (n-1)$ kinds of changes can be generated. In addition, due to the different imaging times of multi-temporal images, the same feature states can be different. For example, the height and abundance of the vegetation canopy in summer and winter are different, which can cause false changes, which interfere with change detection.
2. *Large differences in the scale of urban changes.* Urban objects have the characteristics of multiple scales. For example, in Fig. 1(a), the buildings show different scales in the commercial district, residential area, and industrial area. Urban changes thus inherit the multi-scale characteristics of the objects. Since urban development is a gradual process, the changes are mostly local changes of ground objects. For example, the change from wasteland to road in Fig. 1(b) involves widening the original road and changing part of the area. Compared with building construction, there is a significant difference in scale.

Based on the above mentioned characteristics of urban objects and their changes, in order to better use advanced algorithms to extract urban changes and help with urban sustainable development, different requirements are put forward for remote sensing data. A dataset for use in research into deep learning based change detection algorithms should have the following characteristics:

1. **High spatial resolution.** With the development of space imaging technology, remote sensing Earth observation technology has become more mature. The image acquisition platforms have also gradually transitioned, from satellites that are susceptible to observation conditions to unmanned aerial vehicles (UAVs) which have the advantages of real-time data acquisition and convenient operability, which provides us with the possibility of acquiring high-frequency time-series images. The spatial resolution of the imagery has also gradually improved. According to the resolution, remote sensing images can be divided into four categories, as shown in Fig. 2. Low-to-medium resolution (LTMR) refers to remote sensing data with a spatial resolution of greater than 4 m. Typical satellites include NASA's Landsat 7 and Landsat 8 satellites, and ESA's Sentinel-2 satellite. China's Gaofen-2 satellite is a high-resolution satellite, with a spatial resolution of 1–4 m. Most commercial satellites, such as QuickBird and WorldView, provide very high resolution (VHR) data, with a spatial resolution of 0.3–1 m. Since the development of aerial photogrammetry technology, ultra high resolution (UHR) images can now be obtained through airborne or UAV platforms, for which the spatial resolution is less than 0.3 m (Qin et al., 2016). Urban features have multi-scale characteristics, and many changes are local changes, which can be very fine. From LTMR to UHR data, as the spatial resolution gradually increases, the scale of the visible features gradually decreases, and the details of the objects gradually appear. The higher the spatial resolution of the remote sensing imagery, the richer the spatial detail information it can provide, which allows us to distinguish the characteristics between different objects and obtain accurate change boundaries, providing the possibility for refined UCD. Therefore, to detect changes in the details of objects and solve the problem of the large differences in the scale of urban changes, higher spatial resolution data are required.
2. **Rich prior information of urban object categories.** In the early stage, researchers were keen to study the binary change detection (BCD) task, which is to specify where the change occurred, and to determine whether the object changed. In BCD, we can consider all the land-cover and land-use changes (Nemmour and Chibani, 2006; Liu et al., 2012; Luo et al., 2018), or only pay attention to the changes in a certain type of typical urban object, such as buildings (Jung, 2004; Yang et al., 2018), roads (Zhang, 2004), etc. However, BCD cannot extract the semantic information of the changes, and thus has limitations in practical applications, such as geographic database updating. Nowadays, for the semantic change detection (SCD) task, which is also called multi-class change detection, this involves not only finding the changes but also the change directions, to provide "from-to" information. As a result, SCD is useful for further analysis and application (Huang et al., 2017; Haouas et al., 2019; Daudt et al., 2019). Most SCD methods involve classification, and often use supervised learning models, such as random forest, support vector machine (SVM), or neural network models, which have all achieved good results. Therefore, rich prior information of the urban object categories is needed to obtain the semantic changes and identify complex urban objects.
3. **Large dataset volume.** Massive remote sensing image resources record the ever-changing Earth through long-term observation. In order to utilize this precious information, significant efforts have been made in change detection for remote sensing images (Bruzzone and Prieto, 2000; Du et al., 2002; Liu et al., 2015; Zhu, 2017; Tewkesbury et al., 2015; Zhang et al., 2016; Asokan and Anitha, 2019; Wang et al., 2019b; Ghassemi et al., 2019; Haouas et al., 2019). Change detection methods have also grown apace with the advent of the era of remote sensing big data. Deep learning, as the most efficient way to deal with big data, has been used for change detection (Yang et al., 2019; Yan et al., 2019; Li et al., 2020a; Zhang et al., 2020a; Zheng et al., 2022). Compared with traditional change detection methods, the strong feature extraction and representation capabilities of deep learning mean that it has great potential for UCD. The effect of deep learning methods depends on the volume of the training data. Benefiting from the large-scale publicly available natural image datasets, such as ImageNet (Jia Deng et al., 2009) and COCO (Tsung-Yi Lin et al., 2014), deep learning has achieved great success in computer vision tasks (Krizhevsky et al., 2017; He et al., 2016; Girshick, 2015; He et al., 2017; Perez-Rua et al., 2020; Xie et al., 2020). Large-scale datasets can increase the diversity of the ground objects and the complexity of the changes, and can thus effectively test the generalizability and scalability of change detection methods. Therefore, a larger dataset is needed to promote the research into deep learning algorithms in the field of change detection.

A dataset with all of the above characteristics would allow us to identify complex features and extract semantic changes, based on the powerful feature extraction capability of deep learning, to achieve fine detection of urban changes. However, there are still no public change detection datasets that can meet these requirements, especially for

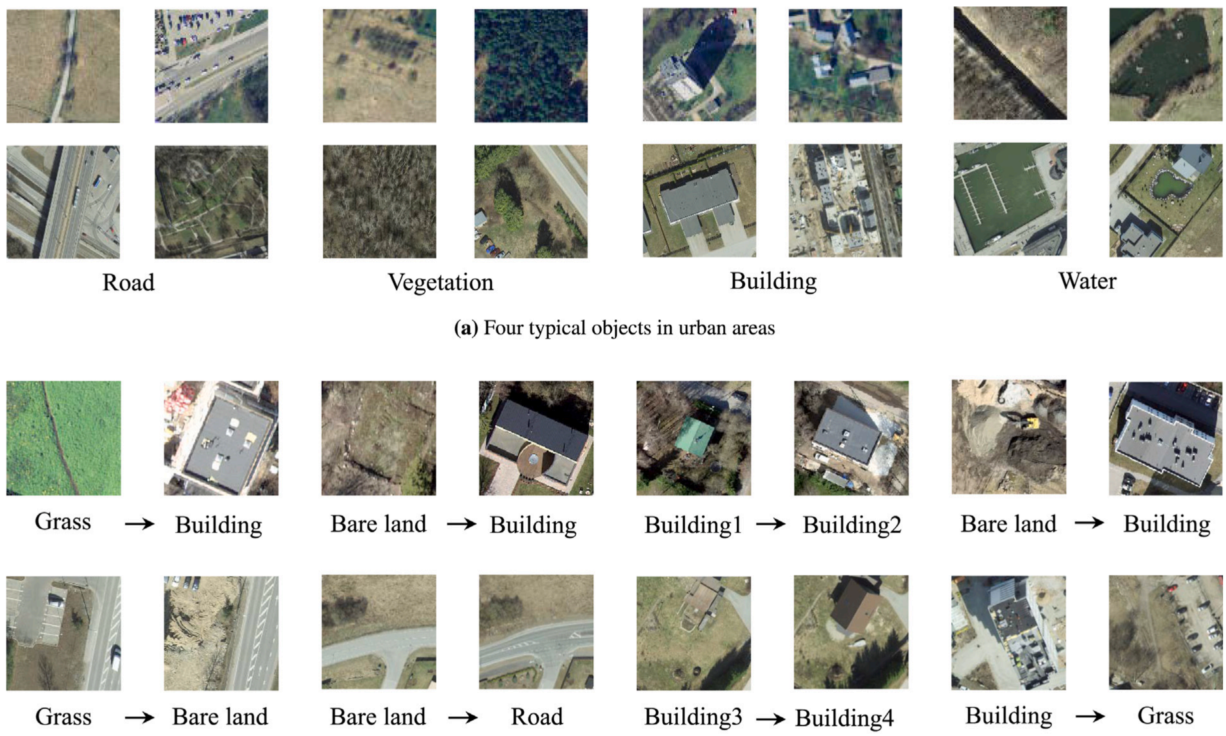


Fig. 1. Examples of urban objects and their changes.

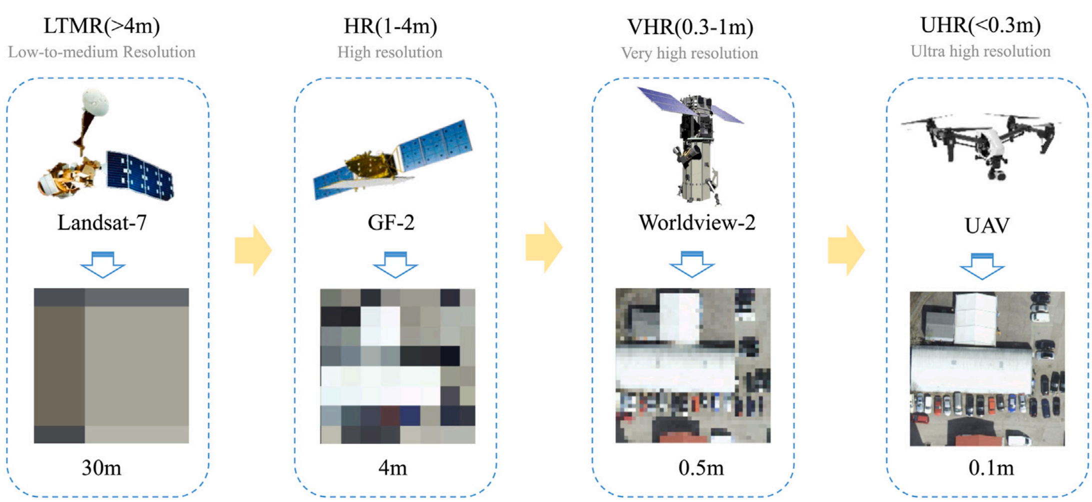


Fig. 2. Examples of images with different spatial resolutions from LTMR to UHR. As the spatial resolution increases, the scale of the visible objects gradually decreases.

SCD. Most of the existing public datasets have some shortcomings. For example, the spatial resolution of the imagery is low, the number of training image pairs is small, or there is a lack of variety of land-cover change for refined UCD. These limitations greatly hinder the development of deep learning based change detection methods for UCD.

In order to address the aforementioned problems, we built a large-scale benchmark dataset for semantic UCD in remote sensing imagery, which we are making publicly available. The dataset is termed the ultra-High Urban Change Detection (Hi-UCD) dataset. This dataset is an extension of our previous work in Tian et al. (2020). We selected an area of 102 km^2 in Tallinn, the capital of Estonia, as the study area to obtain a total of 40 800 image pairs. We would like to highlight three key characteristics of the Hi-UCD dataset, compared to the existing change detection datasets. Firstly, the numbers of images, land-cover classes, and semantic instances are greatly increased. Secondly, the remote sensing images in the Hi-UCD dataset have a spatial resolution of 0.1 m, which can provide rich spatial texture information, but also brings challenges to change detection. Thirdly, after comprehensively considering the typical urban geographic objects and change-related objects, we developed a semantic labeling system, including nine types of land cover and 48 types of semantic change.

In summary, the main contributions of this article are as follows:

1. **The large-scale urban semantic change detection benchmark dataset.** The Hi-UCD dataset, to the best of our knowledge, has the largest number of both land-cover classes and semantic change classes among the benchmark change detection datasets. Meanwhile, it also has a very large number of image

pairs. This dataset will enable the community to develop and validate deep learning based change detection methods. Furthermore, with the ultra-high spatial resolution, it will benefit the detection of more refined urban changes.
2. **Deep learning change detection performance benchmarking.** We benchmarked several representative deep learning based methods in both BCD and SCD on the Hi-UCD dataset, to provide an overview of the performance. In addition, not only did we analyze the relationship between the performance and the characteristics of the different methods, but we also analyzed the effectiveness of different deep learning models for UCD.
3. **New metrics for the evaluation of semantic consistency.** As all we know, changed areas have different classes, while unchanged areas have the same class. Based on this phenomenon, we propose a series of metrics for the evaluation of the semantic consistency in the changed and unchanged regions, from loose to tight. The use of these metrics will allow methods to be evaluated more fairly and comprehensively.
4. **A comprehensive investigation of the challenges of the Hi-UCD dataset when using deep learning change detection methods.** The Hi-UCD dataset is made up of UHR images, and has serious sample imbalance. In order to promote the study of UCD, a series of experiments were conducted to investigate the role of unchanged samples in UCD and the deep learning methods of feature extraction for UHR imagery. After analyzing the experimental results, we found that the unchanged samples are helpful for change detection, and high-resolution networks (HRNet) (Wang et al., 2019a) perform very well in high-resolution feature representation for UCD.

The rest of this article is organized as follows. In Section 2, we review the representative change detection methods, from the traditional handcrafted feature based methods to deep learning feature based methods, and also the public datasets, for both BCD and SCD. In Section 3, we describe the Hi-UCD dataset in detail. In Section 4, we describe how several representative deep learning methods were selected as benchmarks for the Hi-UCD dataset in BCD and SCD. A comprehensive analysis is also provide after comparing the benchmarks. In Section 5, we discuss the challenges in deep learning based SCD with the Hi-UCD dataset. Finally, Section 6 concludes the article.

2. Related work

2.1. Change detection methods

Remote sensing images reflect the color, shape, and spatial geometric characteristics of ground objects and their changes. How to extract the features of the different ground objects and their changes from multi-temporal remote sensing images has become the key to change detection methods. According to the feature extractor, these methods can be divided into handcrafted feature based methods and deep learning feature based methods.

2.1.1. Handcrafted feature based change detection methods

The handcrafted feature based methods take advantage of the obvious gray-scale changes of the ground objects in the images of different periods to extract the changes (Malila, 1980; Nielsen et al., 1998; Nielsen, 2007). With the advent of machine learning, change detection methods began to use more prior information to improve the feature extraction capabilities. As a representative machine learning method, SVM has performed well in change detection tasks as a classifier (Nemmour and Chibani, 2006; Negri et al., 2020). The most commonly used classification based change detection method is post-classification, which is more general in SCD (Afrasinei et al., 2017; Yan et al., 2019). Another classification based method is direct classification, where the multi-temporal images are stacked together as the input of the model. However, the change detection accuracy of these methods is highly related to the classification result, good or not. In order to consider the spatial context, object-based image analysis (OBIA) has been introduced to change detection. Taking geographic objects as the smallest unit of the change detection, and considering the spectral, timing, shape, geometry, spatial, and geographic relationships among all the pixels representing the objects, rich features can be extracted for a comprehensive comparison and analysis (Walter, 2004; Zhang et al., 2017; Wang et al., 2018; Lv et al., 2020). However, the traditional handcrafted feature based methods cannot easily fit non-linear distributions, and a large amount of expert knowledge is required to design a good feature extractor. With the advent of the era of remote sensing big data, the time-consuming traditional methods cannot meet the needs of large-scale change detection applications. At this time, the boom in deep learning has brought another solution to the task of change detection.

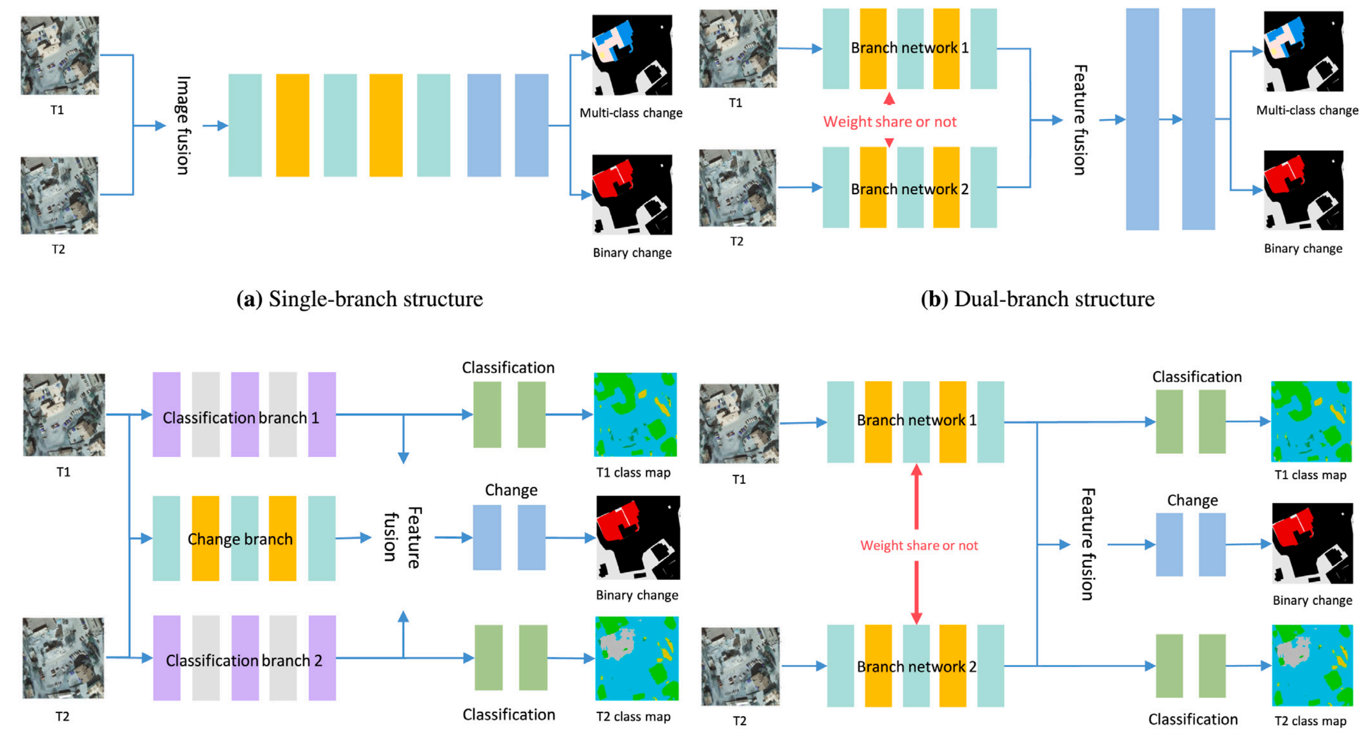
2.1.2. Deep learning feature based change detection methods

Due to the capability of extracting discriminative features and processing massive data, deep learning has made great progress in many domains, especially computer vision (Deng et al., 2009; Liu et al., 2017) and nature language processing (NLP) (Vaswani et al., 2017; Torfi et al., 2020). Earth vision, as a part of computer vision, also has numerous methods developed based on deep learning, which have been applied in remote sensing tasks, including image classification (Hong et al., 2021a,b, 2022), land-cover mapping (Hermosilla et al., 2022; Zhang et al., 2020c; Ma et al., 2021), road extraction (Zhou et al., 2018; Lu et al., 2019; Chen et al., 2021), and also change detection (Chalapathy and Chawla, 2019; Shi et al., 2020).

The change detection methods based on deep learning have been gradually enriched after years of development (Shi et al., 2020). These methods rely on a model incorporating a convolutional neural network (CNN) (LeCun et al., 1998), recurrent neural network (RNN) (Yu et al., 2019), deep belief network (DBN) (Hinton et al., 2006), or autoencoder (AE) (Vincent et al., 2010) to learn spectral-spatial–temporal features from the imagery. According to the structure of the model, deep learning based methods can be categorized into the types shown in Fig. 3.

Single-branch structure. In this structure, as shown in Fig. 3(a), the images are always fused by concatenation, differencing, ratioing, or other methods, to obtain the difference images, which are fed into the deep learning network to extract the deep features of the changes. We can consider the step of image fusion to be preprocessing to obtain the difference images, which highlight some of the information about the changes to guide the network to learn features that can assist in judging where the change happened. Although they can introduce noise or lose some important features, because they act directly on the images, the advanced networks in computer vision can be used, such as DeepLab V3 (Chen et al., 2017), DeepLab V3+ (Chen et al., 2018), U-Net (Ronneberger et al., 2015), etc. For example, Peng et al. (2019) improved U-Net++ as a single-structure network to detect binary change, and Papadomanolaki et al. (2019) added a fully convolutional long short-term memory (LSTM) module to U-Net for temporal modeling. In another way, we can use this single-branch structure to separately classify images from different periods, and then compare the results to obtain the changes (Daudt et al., 2019; Tian et al., 2020). It is a good idea to avoid the influence of image fusion and generate the semantic changes, but the temporal correlation between images of different dates is ignored.

Dual-branch structure. As shown in Fig. 3(b), dual-branch networks are used to separately extract deep features in binary temporal images. Feature fusion is adopted to highlight the features about change, which can be an explicit representation of the change. After this, the deep learning network further explores useful features related to change until the change map is obtained. The operation usually used in feature fusion is concatenation and calculating the $L1$ norm or $L2$ norm. The dual-branch structure can simultaneously extract the deep



(a) Single-branch structure **(b)** Dual-branch structure

(c) Multi-task structure 1 **(d)** Multi-task structure 2

Fig. 3. According to the model structure, deep learning based change detection methods can be divided into three types: (a) single branch structure; (b) dual-branch structure; and (c)–(d) multi-task structure.

features of the input two-period data. If the weight of the dual-branch networks is not shared, the network structure will be a pseudo-Siamese network. If the dual-branch networks share the network weight, it will be a Siamese network that realizes the feature interaction between the images from different periods, which can enhance the network recognition ability and reduce the amount of parameters. Therefore, many dual-branch structure deep learning algorithms are based on a Siamese network. Daudt et al. (2018a) compared three networks to detect binary change in an urban area. The fully convolutional early fusion (FC-EF) model has a single-branch structure, and the images are fused by concatenation in the early stage. The FC-Siam-conc and FC-Siam-diff models are dual-branch structures with a Siamese network and different feature fusion approaches. Chen et al. (2020) constructed a dual-branch structure network with a deep Siamese CNN and a multiple-layer RNN, where the multiple-layer RNN fuses the spatial–spectral features extracted by the deep Siamese CNN, and mines the changes between bitemporal images. In Zhang et al. (2020a), a deep Siamese image fusion network (DSIFN) was proposed, which conducts the feature fusion by concatenation.

Multi-task structure. In order to realize SCD, multi-task structures have also been proposed. BCD can be used to find the changes, and classification can be used to obtain the class information of the ground objects in the different periods. These operations can then be combined to learn more information about the changes, to promote each other's performance. There are two multi-task structures shown in Figs. 3(c) and 3(d). For multi-task structure 1, there are three branches to extract the features separately for the specific task, and they can share the weight of the networks, or not. Based on multi-task structure 1, Daudt et al. (2019) first proposed two networks with separate or integrated strategies to conduct classification and BCD. Yang et al. (2021) also proposed an asymmetric Siamese network to reduce the categorical ambiguity. For multi-task structure 2, the feature extractors for the different tasks are integrated by a Siamese, pseudo-Siamese, or single-branch network to obtain general semantic features, which are more representative, and are optimized by a multi-task loss function. The different network heads then mine task-specific features to generate the class map and change map. Papadomanolaki et al. (2020) constructed a multi-task network based on the model proposed in Papadomanolaki et al. (2019) to generate a building change map and segmentation maps in an urban area. Sun et al. (2020) also adopted multi-task structure 2 with a single-branch to share features in the encoder. It has been shown that BCD and classification by the use of these two multi-task structures can result in an improved accuracy (Daudt et al., 2019; Zheng et al., 2022).

The deep learning feature based methods have been shown to have great potential in UCD. In addition to the structure, the different feature extraction modules and the selection of suitable hyperparameters are also important for fitting the characteristics of change detection in remote sensing imagery.

2.2. Change detection datasets made up of optical remote sensing images

There are several public UCD datasets made up of optical remote sensing images, which have been successfully used in studies. We grouped the public UCD datasets through the spatial resolution of the imagery, and list some of their characteristics in Table 1. We focus on some of the representative BCD and SCD datasets, which are briefly introduced as follows.

2.2.1. Binary change detection datasets

Onera Satellite Change Detection (OSCD) dataset (Daudt et al., 2018b). This dataset focuses on urban areas, and its label changes only include changes caused by urban expansion, while some natural changes (such as vegetation growth or coastline changes) are ignored. The dataset uses Sentinel-2 images, with a uniform resolution of 10 m, and provides single-band, color, and corresponding multi-spectral images. In total, 24 cities were selected as the study area at a global scale,

Table 1
The public datasets for urban change detection with optical remote sensing imagery.

Dataset		Resolution (m)	Image pairs	Image size (pixels)	Change type	Objects of interest	Classes
LTMR	OSCD (Daudt et al., 2018b)	10	24	600 × 600	Binary	Land cover	–
	S2MLTCP (Leenstra et al., 2021)	10	1520	600 × 600	Binary	Land cover	–
	ZY3 (Zhang and Shi, 2020)	5.8	1	1154 × 740	Binary	Land cover	–
HR	SZTAKI AirChange (Benedek and Sziranyi, 2009)	1.5	13	952 × 640	Binary	Land cover	–
	WV3 (Zhang and Shi, 2020)	2	2	1431 × 1431	Binary	Land cover	–
	QB (Zhang and Shi, 2020)	2.4	1	1154 × 740	Binary	Land cover	–
VHR	AICD (Bourdis et al., 2011)	0.5	1000	800 × 600	Binary	Land cover	–
	ABCD (Fujita et al., 2017)	0.4	8506/8444	160 × 160/120 × 120	Binary	Buildings	–
	HRSCD (Daudt et al., 2019)	0.5	291	10000 × 10000	Semantic	Land cover	5
	SECOND (Yang et al., 2021)	0.5–3	4662	512 × 512	Semantic	Land cover	6
	LEVIR-CD (Chen and Shi, 2020)	0.5	637	1024 × 1024	Binary	Buildings	–
	Google Data Set (Peng et al., 2020)	0.55	19	1006 × 1168 – 4936 × 5224	Binary	Buildings	–
	Season-varying (Lebedev et al., 2018)	0.03–1	16 000	256 × 256	Binary	Land cover	–
	xDB (Gupta et al., 2019b)	<0.8	11 034	1024 × 1024	Semantic	Buildings	1
	BDD (Adriano et al., 2021)	1.0	1147	250 × 250	Semantic	Buildings	1
UHR	WHU building (Ji et al., 2019b)	0.2	1	32207 × 15354	Binary	Buildings	–
	Hi-UCD mini (Tian et al., 2020)	0.1	1293	1024 × 1024	Semantic	Land cover	9
	Hi-UCD (proposed)	0.1	40 800	512 × 512	Semantic	Land cover	9

and 24 pairs of images with a size of about 600 × 600 were obtained, with each image containing 13 bands.

SZTAKI AirChange Benchmark Set (Benedek and Sziranyi, 2009). There are three test sets in this dataset, i.e., SZADA, TISZADOB and ARCHIVE, and 13 pairs of 952 × 640 aerial images. Each pair of images is made up of dual-temporal images and their change labels. New buildings, buildings under construction, large areas of vegetation growth, new cultivated land, and building foundations are considered as relevant changes.

High-Resolution Semantic Change Detection (HRSCD) dataset (Daudt et al., 2019). This dataset contains 291 aerial images of 10 000 × 10 000, with a resolution of 0.5 m. The imagery comes from the French National Institute of Geographical and Forest Information, and covers parts of the urban and rural areas of Rennes and Caen in France. HRSCD is the first SCD dataset, and contains a total of five feature categories: water, wet-lands, forests, agricultural areas, and artificial surfaces. Due to the rough classification system, 99.232% of the dataset is "unchanged". The most serious problem is the imbalance of the samples between changed and "unchanged".

LEVIR-CD dataset (Chen and Shi, 2020). This dataset contains 637 pairs of VHR images with a size of 1024 × 1024 and a time span of 5–14 years. In this dataset, the focus is on changes in buildings, including villas, high-rise apartments, small garages, and large warehouses. There are a total of 31 333 building instances. These images are mainly from 20 cities in Texas, USA, and were acquired from 2002 to 2018.

WHU building change dataset (Ji et al., 2019a). This dataset covers 20.5 km^2 and a total of 12 796 buildings. A magnitude 6.3 earthquake occurred in the study area in February 2011 and reconstruction was carried out over the following years. The image acquisition dates for this dataset are 2012 and 2016, during which time a large number of building changes occurred.

2.2.2. Semantic change detection datasets

xDB dataset (Gupta et al., 2019a). This dataset is a large-scale dataset for the advancement of change detection and building damage assessment. The xDB dataset covers 19 disasters around the world, and provides the pre- and post-disaster satellite imagery and building annotations. The xBD dataset contains 11 034 VHR bitemporal optical satellite image pairs with the size of 1024 × 1024, and is designed for locating the buildings in the pre-disaster satellite imagery, and assessing the damage scale (no damage, minor damage, major damage, and destroy) through the post-disaster imagery. Therefore, this dataset is designed for the one-to-many SCD task, i.e., detecting semantic change in one specific type of ground object (Zheng et al., 2021).

Building damage dataset (BDD) (Adriano et al., 2021). The BDD dataset contains multimodal and multi-temporal data for building damage mapping. It contains 1147 image pairs, including pre- and post-disaster satellite imagery and synthetic aperture radar (SAR) data, making it suitable for single-modal, cross-modal and mode fusion scenarios of change detection. Similar to the xDB dataset, it is also suitable for the one-to-many SCD task, because it only requires us to identify the scale of building damage.

SEmantic Change DetectiON (SECOND) dataset (Yang et al., 2021). This dataset uses semantic labeling, and selects six types of objects – ground, trees, low vegetation, water, buildings, and playground – for labeling. The SECOND dataset images come from multiple platforms and multiple sensors. There are 4662 pairs of aerial images from Hangzhou, Shanghai, Chengdu, and other cities in China. The image size is 512 × 512. A total of 30 semantic changes were generated in this dataset.

Hi-UCD mini (Tian et al., 2020). The study area of the Hi-UCD mini dataset is a part of Tallinn, the capital of Estonia, with an area of 30 km^2. Hi-UCD mini contains images from three different years, with each year's images cut into patches with a size of 1024 × 1024. The patches with more than 200 changed pixels have been filtered out. There are 359 image pairs for 2017–2018, 386 pairs for 2018–2019, and 548 pairs for 2017–2019, including images, semantic maps, and change maps at different times.

2.2.3. Limitations of the existing datasets

Through comparison, we found that these datasets all have some limitations, which make them inappropriate for detecting refined urban changes by deep learning methods:

1. **The small dataset scale.** The small scale of the public datasets is not enough for deep learning methods to learn useful features, which results in overfitting and reduces the transferability.
2. **The lack of ultra-high spatial resolution images.** The spatial resolutions of the OSCD, ZY3 (Zhang and Shi, 2020), and SZTAKI AirChange Benchmark Set (Benedek and Sziranyi, 2009) datasets are 10 m, 5.8 m, and 1.5 m, respectively. Although the resolution is gradually increasing, it still cannot meet the requirements of UCD, especially for buildings. Other datasets using VHR images can achieve a higher recognition degree, but they do not distinguish the full boundaries of diverse buildings at multiple scales. UHR images provide more clear boundaries for ground objects, and overcome these problems. However, only the WHU building change dataset (Ji et al., 2019b) uses UHR images.
3. **The lack of semantic annotation.** The ABCD (Fujita et al., 2017), LEVIR-CD (Chen and Shi, 2020), and WHU building (Ji et al., 2019b) datasets only label building-related changes, and the Season-varying dataset (Lebedev et al., 2018) directly labels land-cover related changes, but they all lack semantic changes. It is therefore difficult to perform SCD to obtain fine change directions. Although the HRSCD dataset (Daudt et al., 2019) provides the direction of the changes, its labeling accuracy is only 80% to 85%. In addition, the ground objects of the urban area are classified into five categories, which is relatively rough, and the dataset cannot reflect the changes of the typical objects in urban areas. The SECOND dataset (Yang et al., 2021) was also designed for SCD, and it labels six classes of land cover, resulting in 30 common change categories (including no-change). However, only the changed areas are labeled with land-cover categories, and the no-change areas are not. This is harmful for the recognition of ground objects, because of the huge sample loss.

On the whole, there is a lack of large-scale UHR imagery public datasets with semantic labels, which makes it difficult to achieve satisfactory refined detection for UCD. To solve this problem, we expanded the Hi-UCD mini dataset by expanding the study area from the original $30\,\mathrm{km}^2$ to $102\,\mathrm{km}^2$, and we divided the training, validation, and test sets according to different regions, which can increase the geographic isolation. The new dataset is named the Hi-UCD dataset, which can realize comprehensive detection and analysis of urban changes.

3. Hi-UCD: A large-scale ultra high resolution remote sensing imagery urban semantic change detection benchmark dataset for urban application

3.1. Study area

The Hi-UCD dataset focuses on urban changes, and uses UHR images to construct semantic changes and achieve refined change detection. The study area (as shown in Fig. 4) of the Hi-UCD dataset is a part of Tallinn, the capital of Estonia, with an area of $102\,\mathrm{km}^2$. The Estonian Land Board[1] provided the aerial images, which were taken with a Leica ADS100-SH100 airborne digital sensor in 2018 and 2019. Meanwhile, the topographic database[2] for this area is also publicly available. In order to fairly evaluate the performance of different methods and promote deep learning based change detection development, we divided the imagery into three parts, without overlap. The training set covers a $30\,\mathrm{km}^2$ area, as marked in the orange color in Fig. 4 the area rendered in green is the validation set, with an area of $18\,\mathrm{km}^2$, and the blue area is the test set, which covers about $54\,\mathrm{km}^2$ To facilitate the data loading, the images of the corresponding areas were cropped into 512 × 512 image blocks, with a total of 40 800 pairs.

3.2. Annotation procedure

For the Hi-UCD dataset, we obtained the semantic changes by annotating the land-cover classes in the different periods. The annotation procedure followed the post-classification comparison approach, including three main steps, which are described as follows:

1. **Land-cover class definition**. In this step, we should select the land-cover categories of interest for the application or of interest with respect to the test site. We can refer standard land-cover products like GlobeLand30 (Chen and Chen, 2018) or government standard files to determine the number of categories. Then, we can consider two aspects to modify the number of categories: (1) The ground objects in the test area. Land-cover categories that match the ground objects should be retained and others excluded. (2) The recognition ability of remote sensing imagery. HR imagery can recognize more categories than LTMR imagery. For Hi-UCD dataset, we focus on SCD in urban areas. With reference to the official topographic documents provided by the Estonian Land Bureau, we selected six land-cover categories, including typical urban ground objects (water, vegetation, construction, transportation), and change-related objects (bare land, "other") as the first-level categories, which basically cover all the types of urban land cover in Estonia. Although the first-level categories describe the types of objects in general terms, some of them are too crude to detect changes in internal categories. In order to realize refined detection in the urban area with UHR images, we further divided the three categories of vegetation, construction, and transportation. Finally, there are nine land-cover types, including natural objects (water, grassland, woodland, bare land), artificial objects (building, greenhouse, road, bridge), and "other". Descriptions of every class are provided in Table 2, which are related to the ETAK layer names and descriptions (Maa-amet, 2021).
2. **Land-cover class annotation**. In this step, we should ideally annotate each pixel in the temporal images. In order to reduce the workload and time consumption, we can find the public geographic information database, such as OpenStreetMap[3] (OSM) (Chen et al., 2021; Bonafilia et al., 2019) and National Land Cover Database[4] (NLCD) (Li et al., 2020b), as a basic annotation and improve them to meet our task needs. However, if there is no public database suitable for the test site, the annotation procedure is usually conducted by domain experts through visual interpretation from scratch. Those public databases always have some inconsistent labels with the imagery, because they are from different sources. We should carefully check the labels and imagery by visual interpretation to find the errors and correct them. It can greatly improve the annotation quality.
   For Hi-UCD dataset, there is a land-cover dataset in the Estonian Topographic Database (ETD), for which land-cover classes

[1] Orthophoto, Land Board 2018–2019.

[2] Estonian Topographic Database, Land Board 2018–2019.

[3] https://www.openstreetmap.org.

[4] https://www.usgs.gov/centers/eros/science/national-land-cover-database.

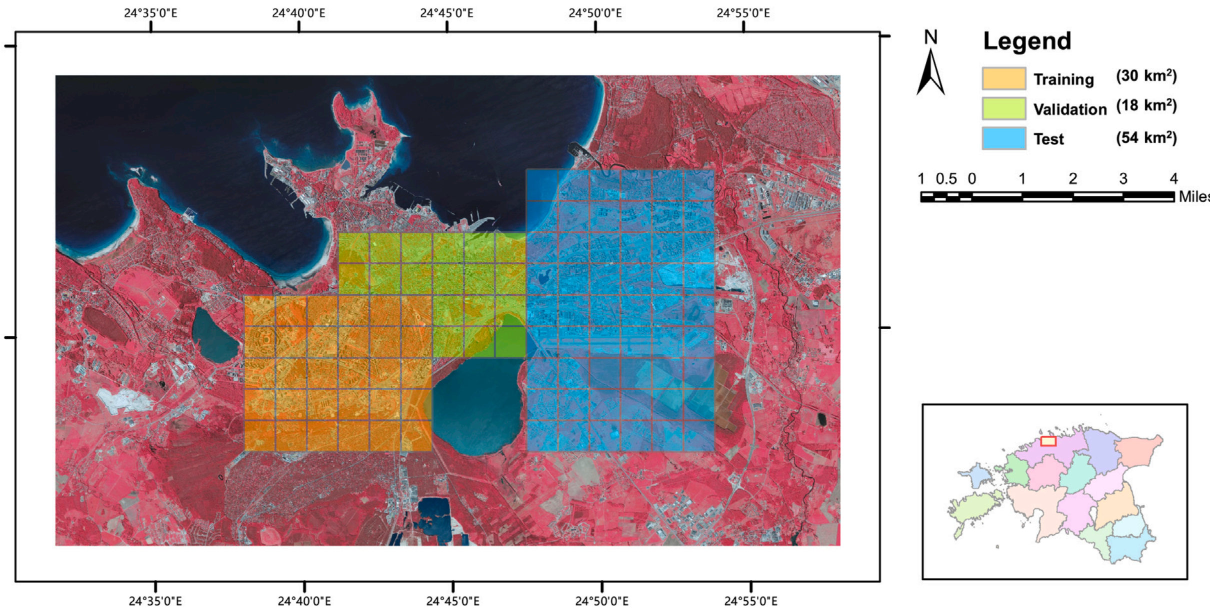


Fig. 4. The study area of the Hi-UCD dataset is a part of Tallinn, the capital of Estonia, with an area of $102\,\mathrm{km}^2$. In order to evaluate the performance of different methods, the dataset is divided into three parts for training, validation, and testing. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
The land-cover classes in the Hi-UCD dataset.

Consideration		Semantic classes	Description
Typical urban objects	Water	Water	Sea. Lake, artificial lake, pond, etc. Watercourse over 8 m wide.
	Vegetation	Grass	Green area, wasteland. Meadow/grassland.
		Woodland	Forest and shrub.
	Construction	Building	Dwelling or public buildings, subsidiary buildings. Production facility, roofed area, other buildings/construction (area > $50\,\mathrm{m}^2$)
		Greenhouse	Greenhouses.
	Transportation	Road	Road area or square.
		Bridge	Bridge or viaduct.
Change related	Bare land	Bare land	Sandy area, stony (pebble) area, open area.
	Other	Other	Building under construction, foundation, ruins. Construction site.

are different from those we defined in step 1. Therefore, the above categories were mapped with the class in EDT through the description in Table 2, obtaining a rough annotation in the different years. We compared the rough annotations provided by the ETD for 2018 and 2019, and found some errors that would affect the change map, especially the inconsistency of the vector boundaries of no-change between different years, and there were also some missing and wrong labels. Since the labels for the buildings had fewer errors, we used the original labels for the buildings. For the other classes, we checked the topographic shape file from 2019 and modified them through visual interpretation to improve the quality. Through visual interpretation, we compared the images of different years to determine the relevant changed objects, and labeled the "other" category. Meanwhile, we checked all the labels again and corrected the errors. After this, two refined land-cover maps were obtained, which are suitable for change detection. In an urban area, the distribution of the ground objects is complex and diverse, and it is not possible to determine the category of each pixel, so we just give the definite annotation for certain area, where it is easy to distinguish the types of the ground object.

3. **Change label generation.** Ideally, comparing the refined land-cover maps can generate the change map. The land cover maps of different times are denoted as LCM_{t_1} and LCM_{t_2}, and the binary change can be formulated by Eq. (1), where $BCD(i,j)$ represents the binary change value of the ith row and jth column.

$$BCD(i,j) = \begin{cases} 0, & LCM_{t_1}(i,j) = LCM_{t_2}(i,j) \\ 1, & LCM_{t_1}(i,j) \neq LCM_{t_2}(i,j) \end{cases} \tag{1}$$

The semantic change transition matrix $M(i,j)$ can be formulated by Eq. (2), where n is the number of land-cover classes.

$$M(i,j) = \{LCM_{t_1}(i,j) * n + LCM_{t_2}(i,j) \mid LCM_{t_1}, LCM_{t_2} \in \{0,1,2,\ldots,n-1\}\} \tag{2}$$

For Hi-UCD dataset, the value of $M(i,j)$ is shown in Table 3, which allows us to look up the related land-cover classes in the different years. In Table 3, the no-change areas ($Area_{NC}$) are collections of pixels numbered as $\{0,10,20,30,40,50,60,70,80\}$, and the others are changed areas ($Area_C$). The code number 20 also represents the change from building to building (i.e., reconstruction) in the Hi-UCD dataset.

Table 3
Semantic change number.

2018	2019								
	Water	Grass	Building	Greenhouse	Road	Bridge	Other	Bare land	Wood land
Water	0	1	2	3	4	5	6	7	8
Grass	9	10	11	12	13	14	15	16	17
Building	18	19	20	21	22	23	24	25	26
Greenhouse	27	28	29	30	31	32	33	34	35
Road	36	37	38	39	40	41	42	43	44
Bridge	45	46	47	48	49	50	51	52	53
Other	54	55	56	57	58	59	60	61	62
Bare land	63	64	65	66	67	68	69	70	71
Wood land	72	73	74	75	76	77	78	79	80

3.3. The characteristics and challenges of the hi-UCD dataset

Compared with the existing UCD datasets listed in Table 1, the Hi-UCD dataset has the following important characteristics:

- **Large size.** The Hi-UCD dataset consists of 102 pairs of optical remote sensing images with the size of 10 000 × 10 000 pixels. We cropped the images into 512 × 512, which is better for deep learning. Finally, we obtained 40 800 pairs of images with semantic labels and changes, of which 12 000 pairs are from the training set, 7200 pairs are from the validation set, and the other 21 600 pairs are from the test set. We manually annotated the images with nine land-cover classes and 48 semantic changes (except for no-change). Compared with the existing datasets, the Hi-UCD dataset has a larger number of land-cover classes and semantic changes.
- **Refined semantic annotation.** We developed the Hi-UCD semantic annotation categories by considering the typical urban objects and change-related objects, including nine types of ground objects. We improved the work efficiency by maximizing the use of the existing topographic database. Meanwhile, through interpretation by experts, the annotation accuracy was also improved. During the annotation procedure, we refined the original labels in the topographic database for each pixel, such as labeling urban greenways and sidewalks in parks as roads, and separated the greenery from the roads. Compared with the other SCD datasets shown in Fig. 5, the Hi-UCD dataset is more conducive to depicting urban areas.
- **Ultra-high spatial resolution.** The Hi-UCD dataset is made up of aerial images with a spatial resolution of 0.1 m, which is the highest resolution among the public datasets. In these images, the geometric shape of the ground objects is clear, and the boundaries are obvious, which provides rich spatial texture information. Therefore, the Hi-UCD dataset is conducive to detecting the local changes of ground objects and realizing refined change detection.

Because of these characteristics, the Hi-UCD dataset provides two special challenges: (1) severe label imbalance; and (2) more noise for change detection.

Severe label imbalance. Due to the large amount of data and the short interval between acquisition dates, of only one year apart, there is serious imbalance among the land-cover types, binary changes, and semantic changes. Fig. 6 shows the number of pixels for each land-cover class in the different years and sets. The distribution of each land-cover class is similar in 2018 and 2019. The numbers of pixels for the greenhouse, bridge and "other" classes show a large gap with the other classes, which will hinder the deep learning models from learning the useful features and making decisions. Apart from this, the classes also have disparity among the different parts of the Hi-UCD dataset, especially the bare land class. Through estimation, the change information obtained from the land cover also has imbalance. The statistics for change and no-change in the training set, validation set, and test set are shown in Fig. 7. Regardless of whether it is considered from pixel or patch, there are few change samples, which leads to severe imbalance between change and no-change. More importantly, the change samples include 48 semantic changes, which also suffer from imbalance. We estimated the number of pixels for each semantic change type in the different sets, and the results are shown in Fig. 8. In the training, validation, and test sets, there are, respectively, 42, 31, and 39 kinds of semantic changes (except for classes that are empty and have only one pixel in the statistics). It is worth noting that, in the different sets, the semantic changes are not always the same, and the training and test sets have their own unique semantic change categories.

More noise for change detection. Change detection can be affected in many ways, such as the radiation differences between images of different time phases or uninteresting changes in the images. In UHR images, these challenges are not diminished, but become even more serious. We show some examples for the Hi-UCD dataset in Fig. 9, which are divided into two groups, according to the reason. Since change detection requires at least two time phase images as input, it is difficult to maintain the same imaging conditions for the different time phase images. Therefore, the first type of challenge is caused by the different imaging conditions, including radiation differences, occlusions, shadows, and the leaning of buildings.

The radiation differences result in an overall shift in the different image spectral values. Due to the difference in solar altitude and imaging angles, it is not only easy to produce occlusions and shadows, but it is also common that different parts of the same object have different view angles (Bruzzone and Bovolo, 2013), especially in urban areas, where there are high-rise buildings and trees. The second type of noise is caused by non-interesting objects. The typical example is seasonal changes, which are just a phenological phenomenon, rather than what we are concerned about in urban change. Due to the high spatial resolution, UHR images increasingly show small-scale objects, and the changes of cars and sundry objects can be seen perfectly clearly. These frequent changes have obvious spectral differences, and can be easily judged as changes. In addition, some changes of buildings are not considered as changes, such as refurbishment of a roof. In general, the UHR images give more detailed information for the ground objects, aggravating the false changes caused by the imaging conditions and non-interesting objects. Therefore, how to use deep learning methods to learn the features from the UHR images, and avoid the influence of the noise has become the key to achieving good results on the Hi-UCD dataset.

Hi-UCD is a novel dataset, with each set having over 70% of the pixels annotated with high accuracy labels. This means that the Hi-UCD dataset can serve for urban SCD, as well as land-cover mapping. Overall, compared with the other existing datasets, the Hi-UCD dataset is far more diverse, comprehensive, and challenging.

4. Experimental results based on deep learning change detection

We benchmarked several representative deep learning based change detection methods for both BCD and SCD on the Hi-UCD dataset, and analyzed their performance in different challenges.

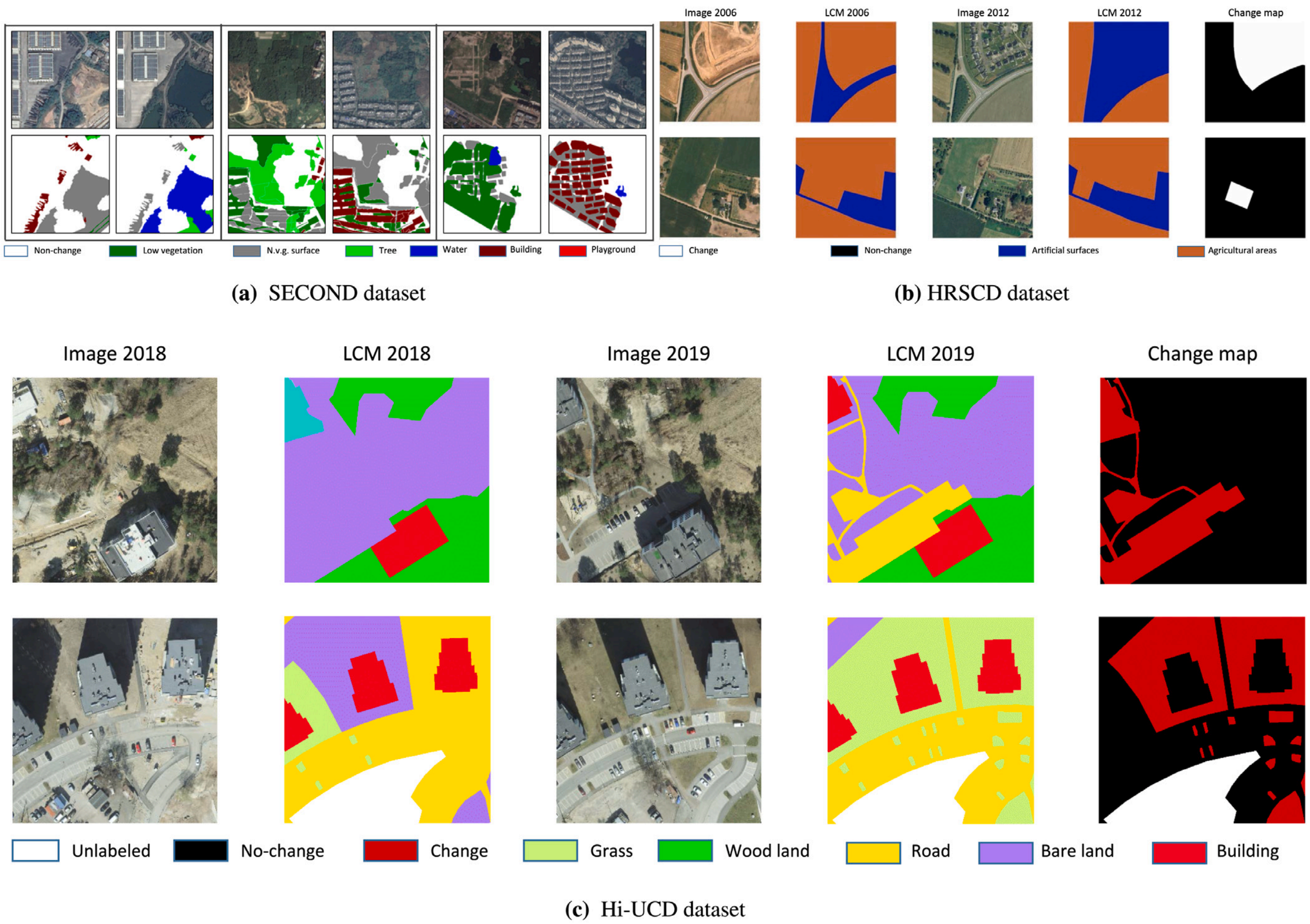


Fig. 5. Comparison of the annotation of the Hi-UCD dataset with that of the existing semantic change detection datasets. The labeling of the Hi-UCD dataset is more precise and accurate.

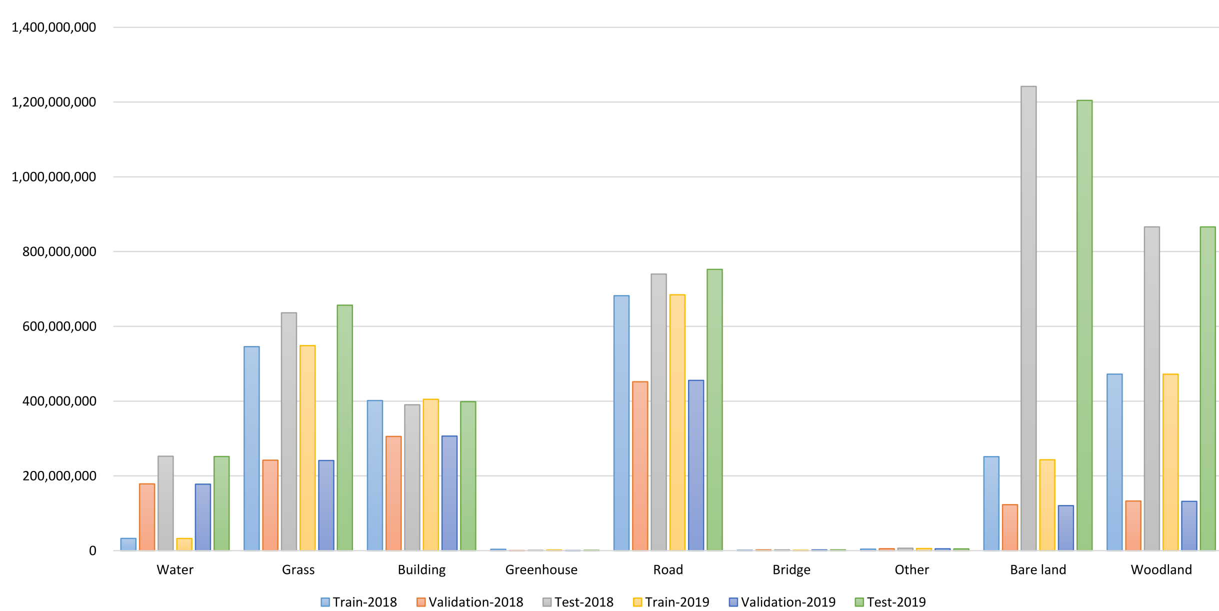
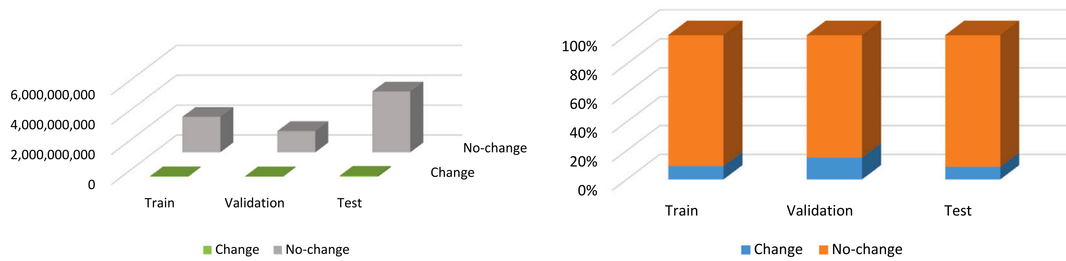
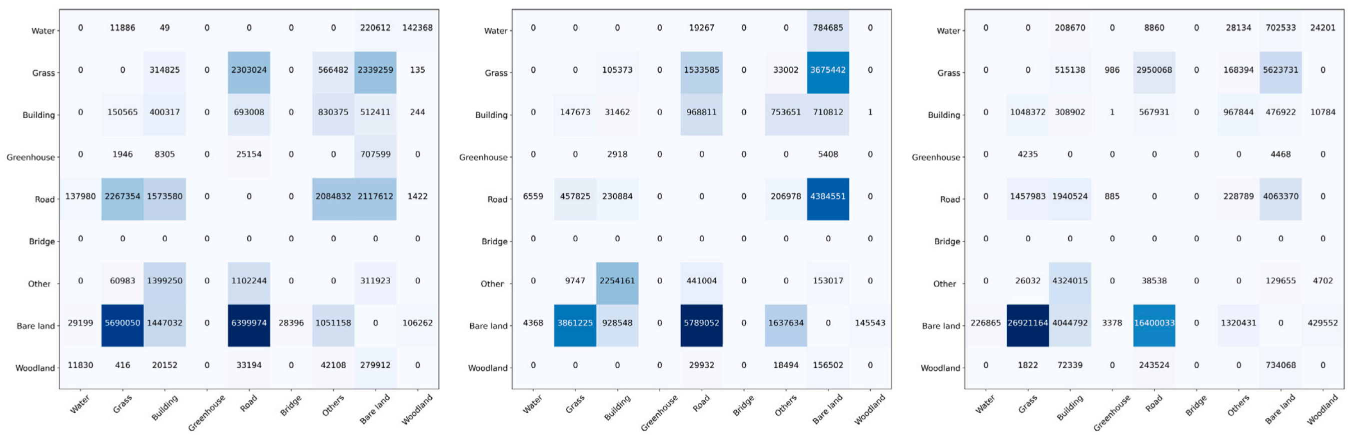


Fig. 6. Statistics for the number of pixels per class.



(a) Statistics for the change and no-change pixels. **(b)** Proportions for the change and no-change patches.

Fig. 7. Statistics for the change and no-change in pixels and patches of the Hi-UCD dataset.



(a) Training **(b)** Validation **(c)** Testing

Fig. 8. Statistics for the semantic changes in pixels of the different sets.

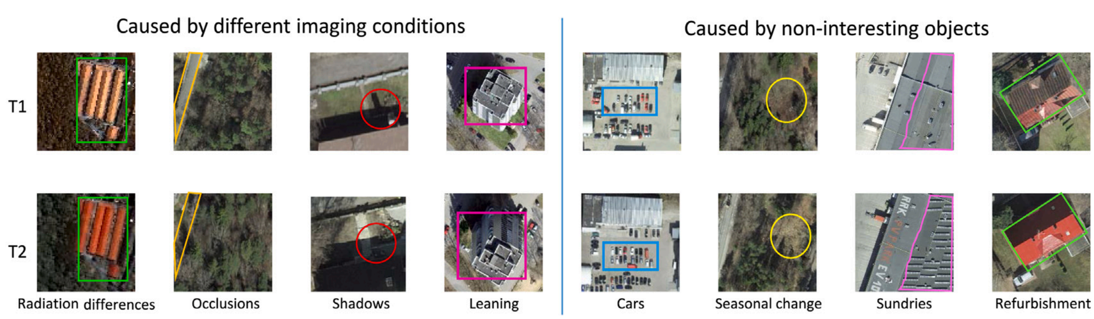


Fig. 9. The noise for change detection with the Hi-UCD dataset.

Table 4
The attributes of the selected benchmark algorithms.

Method		Structure	Fusion	Description	Params (M)	MAdds(B)
Binary change detection	FC-Siam-conc (Daudt et al., 2018a)	Dual-branch	Concatenation	Four layer maxpool and subsampling	1.54	5.28
	FC-Siam-diff (Daudt et al., 2018a)	Dual-branch	Difference	Four layer maxpool and subsampling	1.35	4.67
	FC-EF (Daudt et al., 2018a)	Single-branch	Concatenation	Four layer maxpool and subsampling	1.35	3.54
	FC-Res-EF (Daudt et al., 2019)	Single-branch	–	Adds residual module	1.10	1.99
	U-Net++(Peng et al., 2019)	Single-branch	–	Multiple side-outputs fusion	9.05	33.88
	UNetLSTM (Papadomanolaki et al., 2020)	Single-branch	–	Adds long short-term memory network	8.45	17.3
Sematic change detection	HRSCD Str.1(Daudt et al., 2019)	Single-branch	–	Post-classification	1.10	3.94
	HRSCD Str.2(Daudt et al., 2019)	Single-branch	–	Direct semantic change detection	1.10	2.02
	HRSCD Str.3(Daudt et al., 2019)	Multi-task structure 1	Concatenation	Separate change detection and land-cover mapping	2.20	5.92
	HRSCD Str.4(Daudt et al., 2019)	Multi-task structure 1	Concatenation	Integrated change detection and land-cover mapping	2.31	6.17
	ChangeMask (Zheng et al., 2022)	Multi-task structure 2	Temporal- symmetric transformer	Efficient as backbone	10.62	9.48

4.1. Experimental methods

In order to guarantee that the methods were the same as in the published papers, we only choose methods with open-source code. Finally, a total of 11 representative deep learning based change detection methods, four of which can detect binary change and semantic change simultaneously, were selected as benchmark algorithms. These methods include all of the types of structure and the different types of methods mentioned in Section 2.1.2. In addition to the structural information listed in Table 4, the parameters and the number of operations measured by multiply-adds (MAdds) calculated by a tensor with a size of $1 \times 6 \times 256 \times 256$ are also given to show the deep learning model computational complexity.

FC-Siam-conc, FC-Siam-diff, and FC-EF were proposed by Daudt et al. (2018a), who were the first to use a fully convolutional network (FCN) such as U-Net to deal with the change detection problem. These three algorithms have been widely used as comparison methods in many papers. In Daudt et al. (2019), FC-Res-EF was proposed, which is based on FC-EF with a residual block in both the down-sampling and up-sampling. By utilizing U-Net as a backbone, Papadomanolaki et al. (2019) added LSTM blocks to learn the temporal change pattern. Peng et al. (2019) adopted U-Net++ as a single-branch change detection network to achieve BCD. Meanwhile, deep supervision was conducted with the help of multiple side-outputs fusion in U-Net++. For BCD, deep learning methods always act as the feature extractor for change in multi-temporal imagery. However, deep learning models are sometimes trained for other purposes, such as semantic segmentation or object detection. After this, the semantic change among the ground objects can be generated by comparing the results for different times. Daudt et al. (2019) benchmarked four different SCD strategies on the HRSCD dataset. The first strategy (HRSCD Str.1), performs SCD by post-classification. The second strategy (HRSCD Str.2) directly classifies the semantic change classes, while the third (HRSCD Str.3) and fourth (HRSCD Str.4) strategies are both multi-task structures. HRSCD Str.3 trains two separate networks for land-cover mapping and BCD, as shown in Fig. 3(c), but HRSCD Str.4 integrates these two networks by passing information from the land-cover mapping branches to the change detection branch of the network in the form of difference skip connections. ChangeMask (Zheng et al., 2022) adopts multi-task structure 2 (Fig. 3(d)) as the framework, and adds a temporal-symmetric transformer (TST) to learn a strongly discriminative and temporally symmetric feature representation by spatio-temporal interaction and symmetric fusion.

4.2. Experimental settings and metrics

Settings. We trained the deep learning models on the Hi-UCD training set, and the validation and test set were used for the appraisal. For all the deep learning methods, the learning rate was 0.01, and we used polynomial decay with a decay factor of 0.9. Stochastic gradient descent (SGD) was used for the optimization, with a weight decay of 0.0001 and a momentum of 0.9. For the data augmentation, horizontal and vertical flipping, and rotation of 90 degrees, were adopted during the training. The batch size was 16 and the number of iterations was 50k. We trained the models on a Titan RTX GPU. If a single GPU was out of memory, we used two GPUs. In order to compare the algorithms fairly, we unified the loss function for the same task. The loss functions used for BCD were binary cross-entropy loss and dice loss.

$$\begin{aligned}\mathcal{L}_{bcd} = &-\frac{1}{N}\sum_{N}^{i}(y_i^{bcd}log(p_i^{bcd}) + (1-y_i^{bcd})log(1-p_i^{bcd})) \\ &+ 1 - \frac{2\sum_{i=1}^{N} p_i^{bcd} y_i^{bcd}}{\sum_{i=1}^{N} y_i^{bcd} + \sum_{i=1}}\end{aligned} \tag{3}$$

where y_i^{bcd} is the ground truth for BCD and p_i^{bcd} is the predicted binary probability. The total number of pixels is N. In the classification, we used the cross-entropy loss function $\mathcal{L}_{cls}$. If we have n classes:

$$\mathcal{L}_{cls} = -\frac{1}{n}\sum_{i}^{n} y_i^{cls} log(p_i^{cls}) \tag{4}$$

where y_i^{cls} is the ground truth for the land-cover mapping, and p_i^{cls} is the predicted classification probability. When a multi-task network was used to perform classification and BCD, the loss function used was $\mathcal{L}_{scd} = \mathcal{L}_{bcd} + \mathcal{L}_{cls}^{t_1} + \mathcal{L}_{cls}^{t_2}$.

Binary change detection metrics. We used the overall accuracy (OA) and kappa coefficient to evaluate the overall performance of the BCD results.

$$OA = \frac{1}{N_{bcd}}\sum_{i=1}^{n} x_{ii} \tag{5}$$

$$kappa = \frac{N * \sum_{i=1}^{n_{bcd}} x_{ii} - \sum_{i=1}^{n_{bcd}}(x_{i+} * x_{+i})}{N^2 - \sum_{i=1}^{n_{bcd}} x_{i+} * x_{+i}} \tag{6}$$

where x_{ii} represents the diagonal elements of the confusion matrix; x_{i+} is the sum of the elements in the ith row of the confusion matrix; and x_{+i} is the sum of the elements in the ith column of the confusion matrix. In BCD, n_{bcd} was set to 2. We used the intersection over union (IoU), F1-score, precision, and recall to evaluate the ability to identify change and no-change.

$$precision = \frac{TPs}{TPs + FPs} \tag{7}$$

$$recall = \frac{TPs}{TPs + FNs} \tag{8}$$

$$F_1 = 2 * \frac{precision * recall}{precision + recall} \tag{9}$$

$$IoU = \frac{TPs}{TPs + FNs + FPs} \tag{10}$$

where TPs, FPs, and FNs are the numbers of true positive, false positive, and false negative pixels, respectively.

Semantic change detection metrics. In the SCD task, the multi-task structure has become the mainstream framework of deep learning. Its output results include not only the binary change results from the network and the classification results for the different periods, but also the binary and multi-class change results derived from the above results. The binary change result can be evaluated by Eqs. (5)–(10). The classification result can be evaluated by the IoU_{cls} for every type and the mean intersection over union ($mIoU_{cls} = \frac{1}{n_{cls}} * IoU$) for all the types. In addition, there are multiple types of semantic consistency in multi-change detection under a multi-task structure: (1) *binary consistency*: the consistency between the binary change results obtained by comparing the result of classification for different years (O_{bc}^{cls}) and the binary change results directly output by the network (O_{bc}^{net}); (2) *change area consistency*: the ground object categories in the changed areas of different years are different; (3) *no-change area consistency*: the categories in the unchanged areas of different years are identical to each other. To this end, we designed a series of semantic consistency evaluation indicators, which can more comprehensively evaluate the effectiveness of the various SCD methods.

All the semantic consistency metrics are based on the IoU. The details are as follows:

- *Binary consistency.* By calculating the IoU for the changed and no-change areas between O_{bc}^{cls} and O_{bc}^{net}, $mIoU_{bc}$ is finally obtained as an evaluation of the binary consistency. This process is illustrated in Fig. 10.
- *Change area consistency.* Firstly, the semantic transition result (M_{pred}) of the prediction classification result and the semantic transition (M_{gt}) of the ground truth are calculated by Eq. (2). Then, using O_{bc}^{net} and O_{bc}^{gt}, separately mask M_{pred} and M_{gt} to obtain

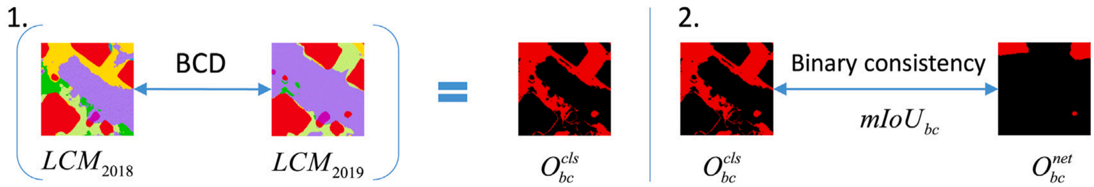


Fig. 10. Binary consistency. Firstly, the binary change result (O_{bc}^{cls}) is obtained by comparing the class maps in different years. The binary consistency between O_{bc}^{cls} and O_{bc}^{net} is then obtained.

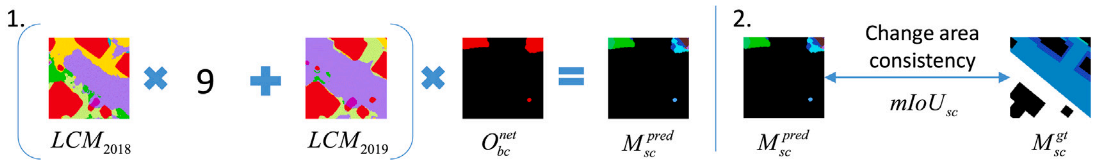


Fig. 11. Change area consistency. Firstly, obtain the semantic transition result in the changed area of prediction (M_{sc}^{pred}) and the ground truth (M_{sc}^{gt}). Then, calculate the change area consistency between M_{sc}^{pred} and M_{sc}^{gt}.

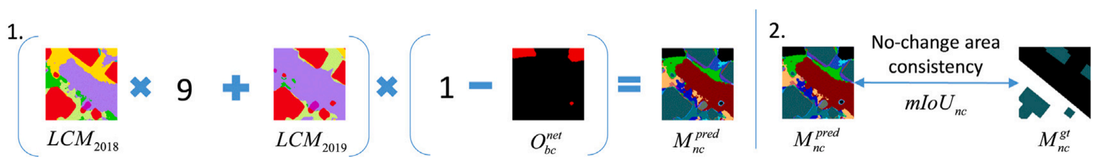


Fig. 12. No-change area consistency. Firstly, obtain the semantic transition result in the no-change area of prediction (M_{nc}^{pred}) and ground truth (M_{nc}^{gt}). Then, calculate the change area consistency between M_{nc}^{pred} and M_{nc}^{gt}.

M_{sc}^{pred}, M_{sc}^{gt}, which treats all the unchanged areas as one category, and calculate the IoU of the n_{change} semantic transition and no-change between M_{sc}^{pred} and M_{sc}^{gt}, finally obtaining $mIoU_{sc} = \frac{1}{n_{change}+1}(\sum_{i=1}^{n_{change}} IoU_i + IoU_{NC})$ as an evaluation of changed area consistency. The IoU_{NC} is the IoU value of the no-change area and n_{change} is the number of semantic changes. The process is illustrated in Fig. 11.

- *No-change area consistency.* After calculating M_{gt} and M_{pred}, using $1 - O_{bc}^{net}$ and $1 - O_{bc}^{gt}$, separately mask M_{pred} and M_{gt} to obtain M_{nc}^{pred}, M_{nc}^{gt}, which treats all changed areas as one category. Calculate the IoU of the $n_{no-change}$ semantic transition and change area between M_{nc}^{pred} and M_{nc}^{gt}, obtaining $mIoU_{nc} = \frac{1}{n_{no-change}+1}(\sum_{i=1}^{n_{no-change}} IoU_i + IoU_C)$ as an evaluation of the unchanged area consistency. The IoU_C is the IoU value of the changed area and $n_{no-change}$ is the number of classes in the no-change area. This process is illustrated in Fig. 12.

4.3. Binary change detection results and analysis

In the visualization example provided for the validation set (Fig. 13), the main changes in this area are the changes between bare land, road, grassland, and the construction and demolition of buildings. When comparing the BCD results of each method with the ground truth, the detection effect for the unchanged areas is better, while that for the changed areas is not satisfactory, and most methods show missed detections. Most of the methods detect the change related to the building in the orange box, which is common in urban areas. In the middle of the image (the magenta box) is the change from bare land to road, where the methods belonging to the single-branch structure perform the best, and the other methods have different amounts of missed detections. In the upper-left corner of the image (the green box), there are buildings which were under construction in 2018 and completed by 2019. FC-Siam-conc, FC-Siam-diff, and HRSCD Str.2 do not completely detect all the parts of this building change, while the other methods detect these changes. However, the boundaries of the changes in these other methods are blurred, and are quite different from the ground truth. In the lower left of the picture (the yellow box), the surrounding of the building has changed from bare land to road, which is covered by shadow. The shadows in the different images cause pseudo-changes, which some methods mistakenly treat as changes. Except for the complete detection of this change by HRSCD Str.1, there are a large number of missed detections in the results of the other methods. However, HRSCD Str.1 works by post-classification, and it shows serious false alarms in the unchanged areas, especially at the boundaries of the ground objects. As a result, the detection effect for the unchanged areas is far worse than the effect of the other methods.

From Table 5, among the six methods that only perform BCD, it can be seen that UNetLSTM performs the best in OA and kappa. Among the indicators evaluated separately for the changed and unchanged areas, the unchanged detection accuracy of all the methods is very good and stable, at more than 95%. This shows that most of the deep learning methods are effective in suppressing false alarms. The various indicators in the changed areas fluctuate greatly. The IoU and F1-score of UNetLSTM are still the best, but each method has its own advantages and disadvantages in precision and recall. FC-EF obtains the highest recall and UNetLSTM obtains the highest precision. The overall performance of the single-branch structure networks is better than that of the dual-branch networks of FC-Siam-conc and FC-Siam-diff. Furthermore, the performance of the multi-task structure SCD methods is usually better than that of the methods using a single-task structure.

For the test set, we selected a scene different from Fig. 13 for display. From Fig. 14, it can be seen that there are more changes between bare land and roads and buildings in this area. The change in the lower right corner (green box) of Fig. 14 is no longer a large block, but is narrow and slender, with a more subtle difference in shape. From the visualization results, all the methods fail in the detail retention, and do not effectively extract the small and narrow road changes. Some changes occurred at the construction site (circled by the orange box), including one change from bare land to grass. As in the magenta box in the validation set, the methods with early fusion patterns detect this change, while the others with a Siamese change branch do not.

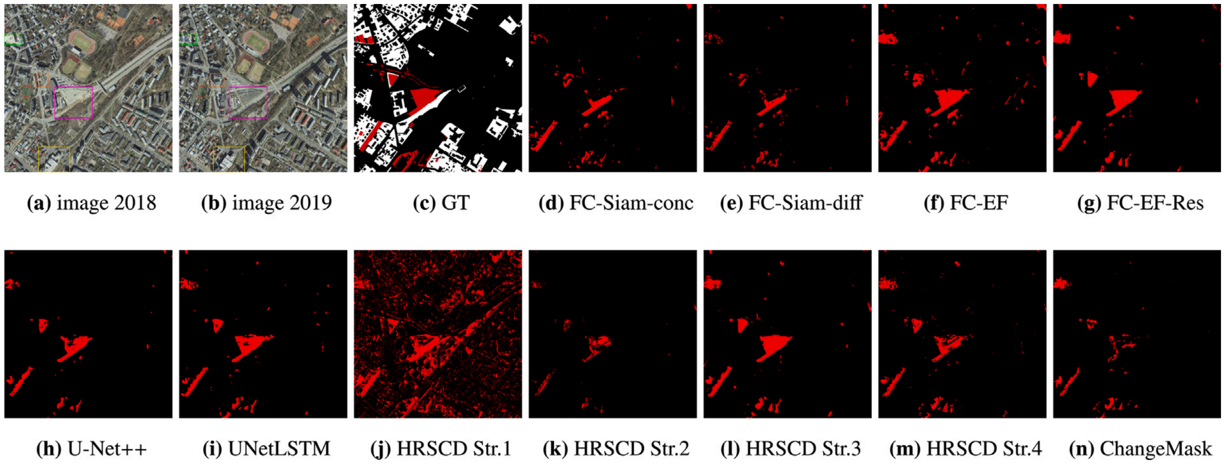


Fig. 13. Visualization of the binary change detection results for the validation set. The image covers 1 km2 with 10,000 × 10,000 pixels, so the reader should zoom in to see the details. Legend: ■ Change, ■ No-change, and white is unlabeled. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Quantitative assessment results for the binary change detection on the validation set. 'NC' represents the 'no-change' class and 'C' represents the change class.

Method	OA (%)	Kappa (%)	*IoU* (%)		F1-score (%)		Precision (%)		Recall (%)	
			NC	C	NC	C	NC	C	NC	C
FC-Siam-conc	97.25	32.32	97.23	20.28	98.60	33.72	98.62	33.29	98.57	34.16
FC-Siam-diff	97.65	34.28	97.64	21.55	98.80	35.46	98.58	40.53	99.03	31.52
FC-EF	96.09	34.49	96.05	22.16	97.98	36.27	99.03	27.23	96.97	54.32
FC-EF-Res	97.50	44.14	97.47	29.37	98.72	45.40	98.97	41.02	98.47	50.83
U-Net++	97.84	39.77	97.82	25.67	98.90	40.86	98.68	46.36	99.12	36.52
UNetLSTM	97.88	45.58	97.86	30.43	98.92	46.66	98.86	48.14	98.98	45.27
HRSCD Str.1	88.04	15.47	87.88	10.16	93.55	18.45	99.21	10.72	88.50	66.09
HRSCD Str.2	98.03	20.15	98.03	11.56	99.00	20.72	98.20	59.15	99.82	12.56
HRSCD Str.3	97.51	43.58	97.49	28.90	98.73	44.84	98.94	41.05	98.52	49.41
HRSCD Str.4	98.09	47.67	98.07	32.13	99.03	48.64	98.84	54.11	99.22	44.17
ChangeMask	98.37	48.53	98.36	32.71	99.17	49.29	98.73	67.92	99.62	38.68

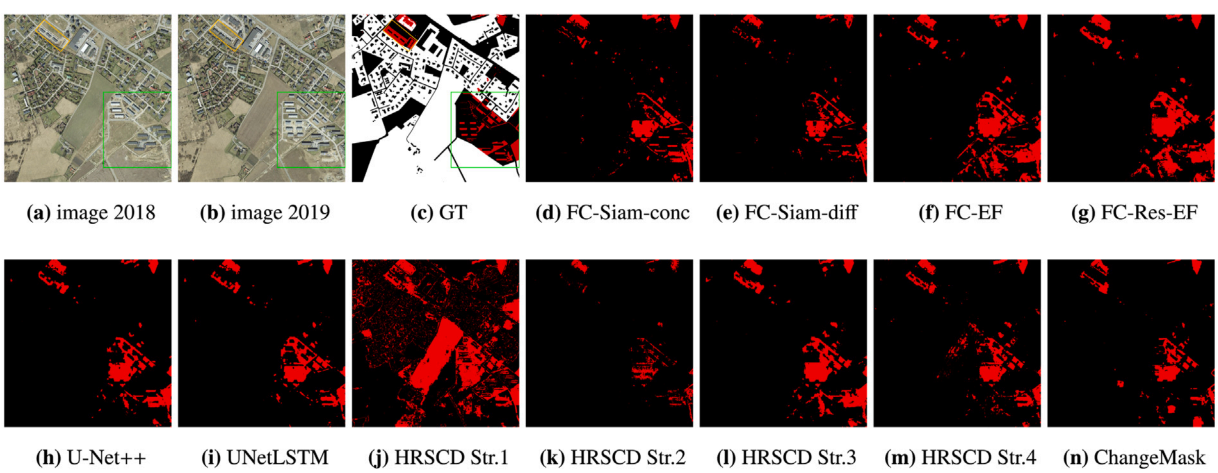


Fig. 14. The binary change detection results for the test set. The image covers 1 km^2 with 10,000 × 10,000 pixels, so the reader should zoom in to see the details. Legend: ■ Change, ■ No-change, and white is unlabeled. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

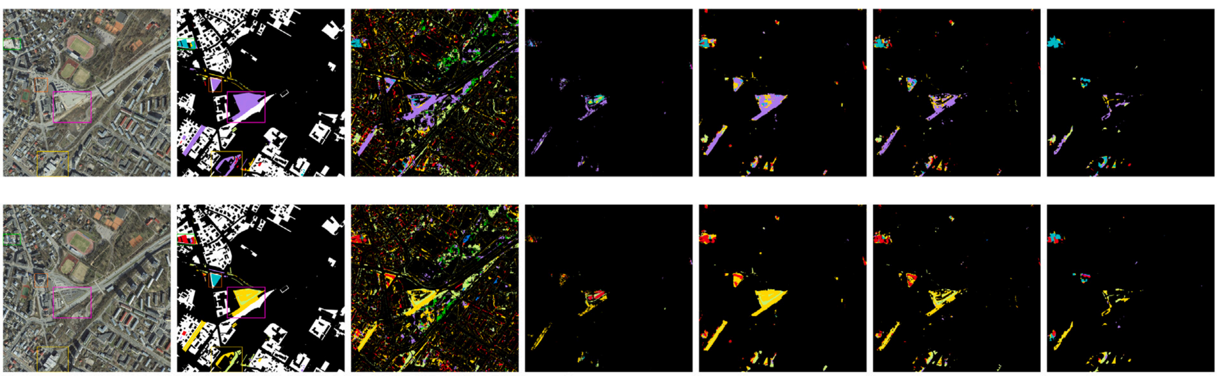
The quantitative assessment result are provided in Table 6, where HRSCD Str.1 obtains the lowest score because its results contain many false alarms. UNetLSTM performs well, because the LSTM network effectively extracts the spatio-temporal features. ChangeMask performs the best, and the IoU score of the changes exceeds that of the other methods by a lot. ChangeMask uses a backbone network with stronger feature extraction capabilities, and has a stable effect on the validation and test sets.

4.4. Semantic change detection results and analysis

The visualization results shown in Figs. 15 and 16 are the land-cover maps of each year, masked by the binary change map to represent semantic change. From the results for SCD in Figs. 15 and 16, the false alarms in the result of HRSCD Str.1 are high, which reduces the overall detection effect. HRSCD Str.2 directly classifies semantic change. Although it shows a lot of improvement over HRSCD Str.1 in

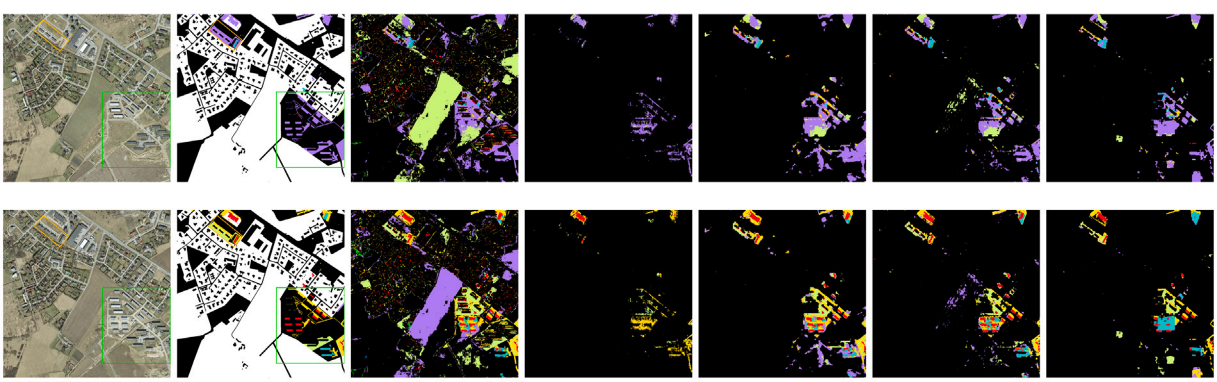
Table 6
Quantitative assessment results for binary change detection on the test set. 'NC' represents the 'no-change' class and 'C' represents the change class.

Methods	OA (%)	Kappa (%)	*IoU* (%)		F1-score (%)		Precision (%)		Recall (%)	
			NC	C	NC	C	NC	C	NC	C
FC-Siam-conc	97.03	30.41	97.01	18.97	98.48	31.90	98.82	27.60	98.14	37.77
FC-Siam-diff	97.82	36.31	97.80	23.02	98.89	37.42	98.79	39.62	98.99	35.45
FC-EF	96.14	31.52	96.10	19.94	98.01	33.24	99.08	24.38	96.96	52.23
FC-EF-Res	96.96	36.31	96.93	23.29	98.44	37.77	99.05	30.33	97.84	50.08
U-Net++	97.57	36.89	97.55	23.55	98.76	38.13	98.88	35.90	98.64	40.65
UNetLSTM	97.82	42.55	97.81	27.92	98.89	43.65	98.98	41.75	98.80	45.74
HRSCD Str.1	84.93	11.97	84.73	8.04	91.73	14.88	99.38	8.30	85.18	71.48
HRSCD Str.2	98.22	21.97	98.22	12.70	99.10	22.54	98.41	56.98	99.80	14.05
HRSCD Str.3	97.09	36.61	97.07	23.48	98.51	38.03	99.02	31.33	98.01	48.38
HRSCD Str.4	98.14	45.52	98.12	30.26	99.05	46.46	98.95	49.41	99.16	43.84
ChangeMask	98.49	47.94	98.48	32.15	99.23	48.66	98.86	64.98	99.61	38.89



(a) images (b) GT (c) HRSCD Str.1 (d) HRSCD Str.2 (e) HRSCD Str.3 (f) HRSCD Str.4 (g) ChangeMask

Fig. 15. The semantic change detection results for the validation set. The top row is the result for 2018, and the bottom row is the result for 2019. The image covers 1 km^2 with 10,000 × 10,000 pixels, so the reader should zoom in to see the details. Legend: Water, Grass, Building, Greenhouse, Road, Bridge, Bare land, Woodland, Other, no-change, and white is unlabeled. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



(a) images (b) GT (c) HRSCD Str.1 (d) HRSCD Str.2 (e) HRSCD Str.3 (f) HRSCD Str.4 (g) ChangeMask

Fig. 16. The semantic change detection results for the test set. The top row is the result for 2018, and the bottom row is the result for 2019. The image covers 1 km^2 with 10,000 × 10,000 pixels, so the reader should zoom in to see the details. Legend: Water, Grass, Building, Greenhouse, Road, Bridge, Bare land, Woodland, Other, no-change, and white is unlabeled. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the issue of unchanged false alarms, the imbalance of samples between the unchanged and changed categories greatly reduces the effect of the SCD. The HRSCD Str.2 model fails to learn effective features to distinguish the different types of semantic changes, resulting in low evaluation scores. Some pseudo-changes caused by vegetation phenology are still difficult to identify, as in the result of HRSCD Str.1 in Fig. 16. The multi-task methods mask the classification results by a binary change map, so that the false alarms caused by post-classification can be eliminated. In terms of the classification results, HRSCD Str.4 performs better than HRSCD Str.3, but ChangeMask obtains the highest score. Generally speaking, the types of objects in the changed areas are different. However, the land-cover types of the changed areas may be classified into the same class, due to classification errors, and the area will be expressed as unchanged. This is the case in the results of HRSCD Str.3, where the changed and unchanged areas in the BCD result are inconsistent with the SCD result. This is due to the inconsistency between the post-classification comparison result produced by the classification branch and the change result directly output by the change branch. Most of the multi-task methods are faced with the problem of semantic inconsistency between the three parts, i.e. binary consistency, change area consistency, and no-change area consistency. This is also reflected in Tables 7 and 8, where the three semantic consistency

Table 7
The quantitative assessment results for the semantic change detection on the validation set.

Model	2018			2019			Semantic consistency		
	$mIoU$	OA	Kappa	$mIoU$	OA	Kappa	$mIoU_{bc}$	$mIoU_{sc}$	$mIoU_{nc}$
HRSCD Str.1	51.11	85.20	81.37	51.00	85.47	81.67	–	8.11	46.16
HRSCD Str.2	–	–	–	–	–	–	–	4.08	–
HRSCD Str.3	50.94	85.18	81.35	51.00	85.36	81.53	49.25	10.30	47.27
HRSCD Str.4	51.34	85.00	81.08	51.82	85.31	81.49	48.63	11.84	47.81
ChangeMask	54.66	86.43	82.92	55.72	86.79	83.35	49.68	13.17	53.97

Table 8
The quantitative assessment results for the semantic change detection on the test set.

Model	2018			2019			Semantic consistency		
	$mIoU$	OA	Kappa	$mIoU$	OA	Kappa	$mIoU_{bc}$	$mIoU_{sc}$	$mIoU_{nc}$
HRSCD Str.1	45.78	80.32	75.28	49.55	82.21	77.84	–	6.08	42.08
HRSCD Str.2	–	–	–	–	–	–	–	3.36	–
HRSCD Str.3	45.94	80.40	75.37	49.71	82.27	77.92	47.13	10.37	43.71
HRSCD Str.4	47.56	80.42	75.45	50.56	82.03	77.64	46.76	11.17	45.76
ChangeMask	55.95	82.77	78.53	56.28	83.07	78.97	47.53	14.48	53.44

scores reflect this phenomenon from binary to semantic change. The existing methods have low semantic consistency scores. Therefore, how to improve the semantic consistency of the multi-task structure methods is a problem that needs to be solved.

For the SCD accuracy assessment results listed in Tables 7 and 8, we respectively evaluated the classification effect and semantic consistency of each method. The HRSCD series of methods have similar scores in classification. ChangeMask performs the best and has better robustness. From the perspective of the various evaluation indicators, HRSCD Str.4 performs slightly better than HRSCD Str.3. This proves that enhancing the connection between the change detection branch and the classification branch is beneficial to the improvement of the change detection accuracy (Daudt et al., 2019). Meanwhile, semantic changes are more fragmented, which requires the models to have the ability to recognize different class boundaries. Fig. 17 shows the evaluation results for each semantic change in the validation and test sets. The detection of semantic changes can be very difficult in Hi-UCD because of the large number and imbalance of categories. Many semantic changes cannot be detected. Obtaining the semantic changes through classification can effectively solve the problem of the different types of semantic changes in the training and test sets, such as the No. 12 semantic change (grass to greenhouse) in the test set.

5. Discussion of the challenges of the Hi-UCD dataset when using deep learning change detection methods

5.1. The role of unchanged samples

In Section 3.3, we mentioned that there is serious sample imbalance in the Hi-UCD dataset. A large number of existing works have shown that too many unchanged samples affect the feature learning for the changed samples. Therefore, we raise a question that is worth discussing: is it necessary to delete the redundant unchanged samples? We conducted experiments on the Hi-UCD dataset to answer this question, and we explored the role of unchanged samples in change detection.

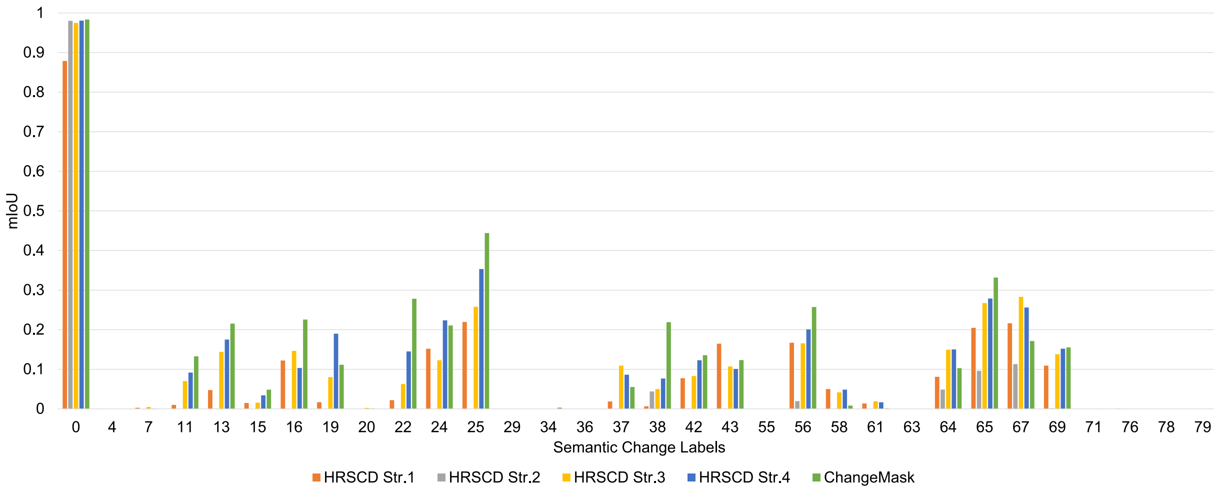
After completely removing the unchanged image patches, there are only 1117 pairs of changed image patches, which were used as the training data. After selection, the changed pixels accounted for approximately 12% of all 1117 pairs of image patches. The number of training iterations was 7500 and the evaluation was conducted on the test set for BCD and SCD. The other settings were the same as in the previous experiments. The quantitative assessment results for the BCD and SCD are listed in Tables 9 and 10.

For the BCD, we compared Table 6 with Table 9. Although the number of unchanged samples has been greatly reduced, and the unchanged accuracy has hardly changed, it is the changed accuracy that is mainly affected. The performance of most of the methods has declined, except for HRSCD Str.2, which has improved a lot. The more model parameters, the greater the decrease in experimental performance. For example, the experimental effect of FC-Siam-conc, FC-Siam-diff, FC-EF, and FC-EF-Res is almost unaffected, while UNetLSTM's kappa drops by 12%. The multi-task structure methods also show a decrease, and HRSCD Str.4 shows a drop of 10%. Differing from the other datasets, Hi-UCD with UHR images results in high complexity for the unchanged areas. As a result, the unchanged samples still provide a large amount of ground object information, helping the model distinguish between change and no-change in BCD, especially for models with more parameters.

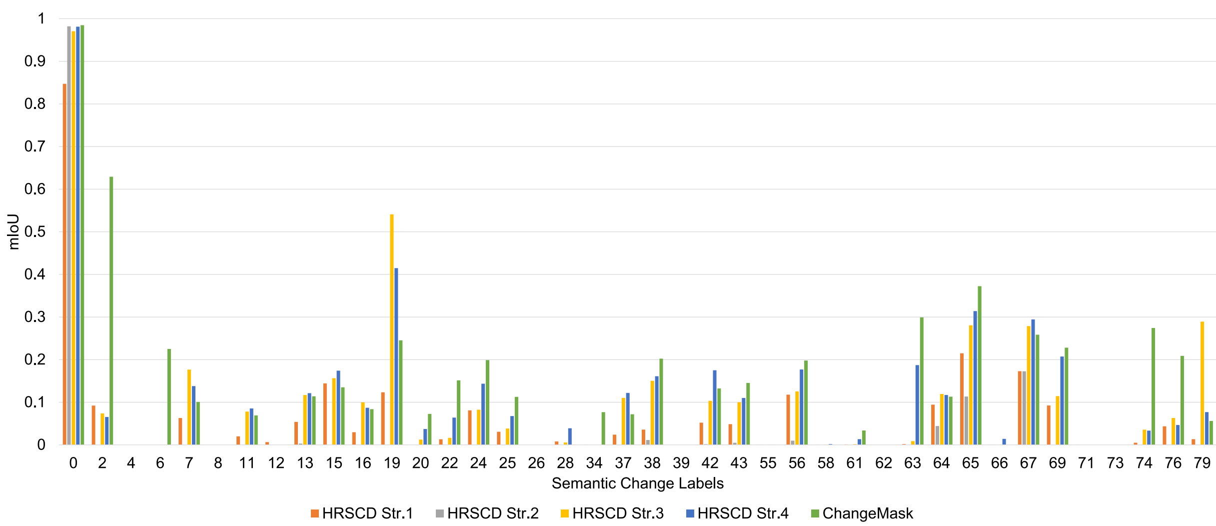
Comparing Table 10 with Table 8, although the accuracy of the multi-task structure method decreases more in BCD, it decreases less in $mIoU_{bc}$ and $mIoU_{sc}$. The increased accuracy of HRSCD Str.2 in BCD and SCD shows that sample imbalance has a greater impact on the direct change detection methods. In addition to the binary change branch, the classification ability is also affected by the performance of the multi-task structure method. Therefore, we summarize the quantitative assessment results of HRSCD Str.1, HRSCD Str.3, HRSCD Str.4, and ChangeMask under different training rules in Table 10. Comparing the classification results in Table 10 with those in Table 8, it can be seen that the classification metrics are reduced by around 6%–12% after removing the no-change image pairs in the training set. These unchanged samples with land-cover labels are beneficial for the classification. Comparing the three semantic consistency metrics, the $mIoU_{nc}$ score is reduced by nearly 8%–12%. Although the reduction in the unchanged samples greatly reduces the accuracy of the BCD and classification score, it has little effect on the $mIoU_{bc}$ and $mIou_{sc}$. This is because, if we preserve the unchanged samples, the classification is improved mostly in the unchanged areas, where the BCD results are always good in a multi-task structure. From the above analysis, it has been shown that unchanged samples can effectively improve the network classification capabilities in unchanged areas, and can assist in identifying changes. How to improve the model's semantic recognition ability in the changed areas is the key to improving the detection performance further.

5.2. Ultra high resolution image feature extraction

The Hi-UCD dataset has an ultra-high spatial resolution, which brings a lot of background noise while providing clear boundaries. At the same time, more attention is paid to the changes in the details of the ground objects when labeling, resulting in many small target changes in the Hi-UCD dataset. We described the challenges caused by



(a) Validation set



(b) Test set

Fig. 17. The accuracy for each semantic change in the validation and test sets.

Table 9
The quantitative assessment results for the binary change detection on the test set, with the model trained by the selected training set. 'NC' represents the 'no-change' class and 'C' represents the change class.

Method	OA (%)	Kappa (%)	*IoU* (%)		F1-score (%)		Precision (%)		Recall (%)	
			NC	C	NC	C	NC	C	NC	C
FC-Siam-conc	96.39	30.17	96.36	18.94	98.15	31.85	98.97	24.43	97.34	45.76
FC-Siam-diff	97.24	35.03	97.21	22.26	98.59	36.41	98.92	31.60	98.26	42.95
FC-EF	96.10	32.04	96.07	20.30	97.99	33.76	99.11	24.58	96.90	53.86
FC-EF-Res	97.15	35.07	97.12	22.31	98.54	36.49	98.95	30.92	98.13	44.49
U-Net++	97.13	35.37	97.11	22.54	98.53	36.78	98.96	30.98	98.11	45.27
UNetLSTM	95.79	30.54	95.75	19.29	97.83	32.34	99.12	22.98	96.57	54.56
HRSCD Str.1	80.52	9.18	80.26	6.52	89.05	12.25	99.39	6.68	80.65	73.74
HRSCD Str.2	97.73	33.24	97.72	20.76	98.85	34.39	98.73	36.78	98.96	32.29
HRSCD Str.3	97.04	34.59	97.02	21.98	98.49	36.04	98.96	29.95	98.02	45.21
HRSCD Str.4	97.24	35.99	97.22	22.97	98.59	37.36	98.95	32.09	98.22	44.71
ChangeMask	97.54	43.61	97.51	28.89	98.74	44.83	99.14	38.18	98.35	54.30

UHR imagery in Section 3.3. The above-mentioned problems require the model to make full use of the detailed expressive ability of the UHR images and suppress the background changes, to effectively identify the boundaries and types of ground objects. The existing methods listed in Table 4 do not use a specific module to extract the features of high-resolution images. In order to solve this problem, we used HRNet (Wang et al., 2019a) pretrained by ImageNet (Deng et al., 2009) as the backbone to perform SCD in accordance with the framework of multi-task structure 2 (Fig. 3(d)). The feature fusion approach used in HRNet is the absolute difference. HRNet was proposed by Wang et al.

Table 10
The quantitative assessment results for the semantic change detection on the test set, with the model trained by the selected training set.

Model	2018			2019			Semantic consistency		
	$mIoU$	OA	Kappa	$mIoU$	OA	Kappa	$mIoU_{bc}$	$mIoU_{sc}$	$mIoU_{nc}$
HRSCD Str.1	39.04	74.20	67.51	42.9	76.06	70.11	–	5.85	34.64
HRSCD Str.2	–	–	–	–	–	–	–	4.44	–
HRSCD Str.3	39.07	74.22	67.54	42.92	76.08	70.14	43.88	9.78	35.96
HRSCD Str.4	39.25	73.77	66.95	42.66	75.29	69.13	44.09	10.27	36.04
ChangeMask	43.62	77.89	72.02	45.66	79.31	74.05	47.01	14.02	41.27

Table 11
The binary change detection results of HRNet on the test set. 'NC' represents the 'no-change' class and 'C' represents the change class.

Method	Params	MAdds	OA (%)	Kappa (%)	IoU (%)		F1-score (%)		Precision (%)		Recall (%)	
					NC	C	NC	C	NC	C	NC	C
ChangeMask	10.62	9.48	98.49	47.94	98.48	32.15	99.23	48.66	98.86	64.98	99.61	38.89
HRNet-W18	10.97	17.71	98.43	48.99	98.42	33.12	99.20	49.76	98.92	60.68	99.49	42.16
HRNet-W32	33.74	49.24	98.49	50.03	98.48	34.02	99.23	50.76	98.92	63.41	99.54	42.33
HRNet-W40	52.45	75.06	98.45	51.61	98.44	35.49	99.21	52.39	99.00	60.28	99.43	46.32
HRNet-W48	75.29	106.56	98.34	49.47	98.32	33.60	99.15	50.30	98.99	55.90	99.32	45.73

Table 12
The quantitative assessment results for the classification and semantic change detection of HRNet on the test set.

Model	Params	MAdds	2018			2019			Semantic consistency		
			$mIoU$	OA	Kappa	$mIoU$	OA	Kappa	$mIoU_{bc}$	$mIoU_{sc}$	$mIoU_{nc}$
ChangeMask	10.62	9.48	55.95	82.77	78.53	56.28	83.07	78.97	47.53	14.48	53.44
HRNet-W18	10.97	17.71	54.74	83.43	79.26	55.92	83.86	79.89	49.10	17.20	51.73
HRNet-W32	33.74	49.24	53.82	82.85	78.46	56.63	84.46	80.59	47.96	17.03	50.65
HRNet-W40	52.45	75.06	53.53	83.26	78.95	54.52	84.47	80.61	48.82	16.93	49.48
HRNet-W48	75.29	106.56	52.89	82.96	78.64	52.78	83.50	79.44	49.30	12.02	49.73

(2019a), and is an approach which can learn a high-resolution representation from imagery, while also having numerous applications in Earth vision (Seong and Choi, 2021; Guo et al., 2021; Osco et al., 2021; Chen et al., 2021). HRNet has different number of layers, which we denote as HRNet-Wx (x ={18, 32, 40, 48}), where x means the network width. The experimental settings of the HRNet semantic change model were the same as those in Section 4.2. Table 11 lists the BCD results of HRNet. The classification and SCD results are listed in Table 12.

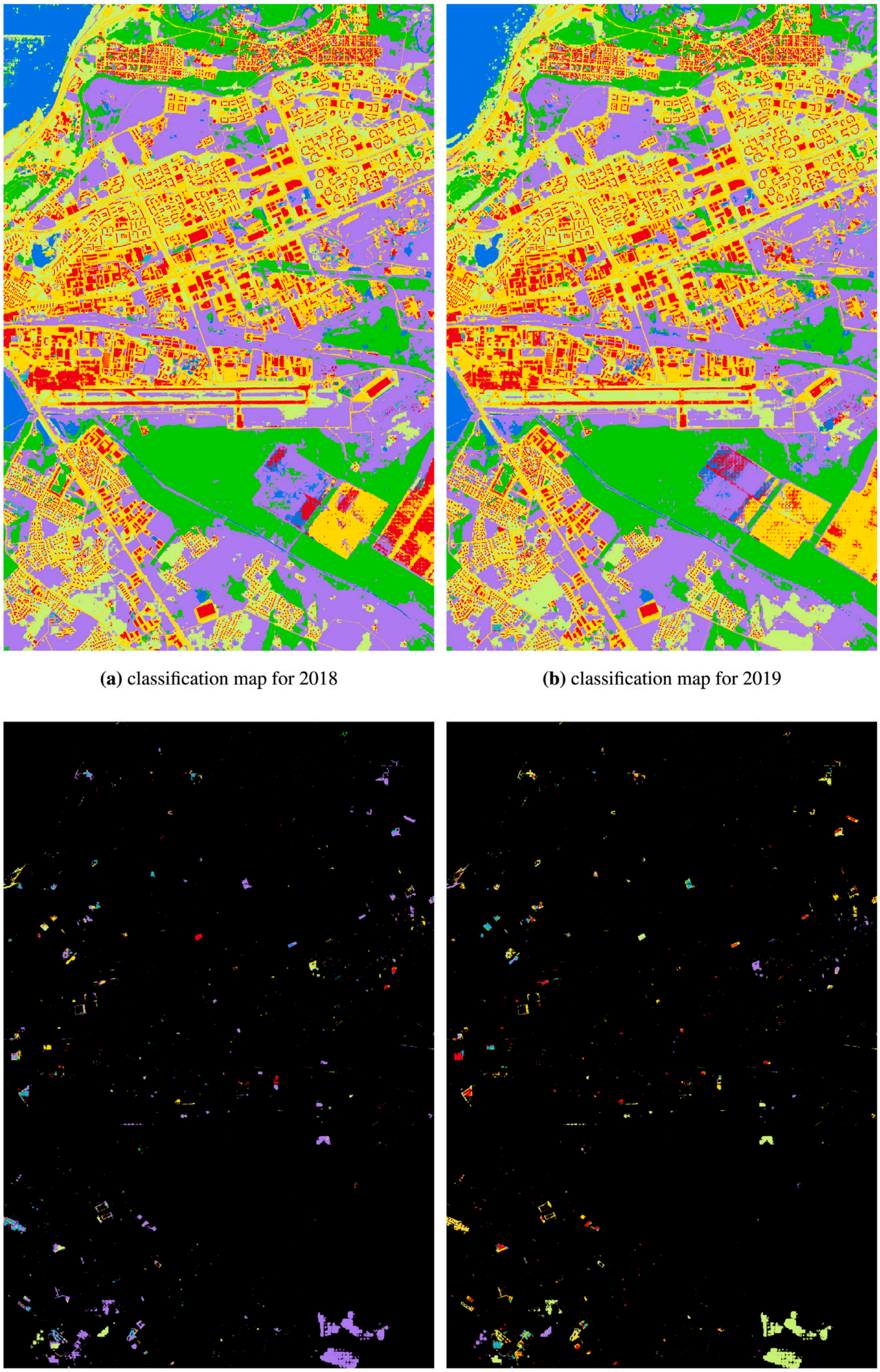
Compared with the ChangeMask method, which obtains the best performance among the existing SCD methods, HRNet shows a superior performance. From the perspective of semantic consistency, HRNet pays more attention to the consistency of the changed areas. The classification results of HRNet have declined compared with ChangeMask, resulting in a decrease of 2%–4% for the no-change consistency. HRNet-W40 performs the best in BCD, and HRNet-W18 performs the best in SCD. Overall, HRNet-W18 is more robust on the Hi-UCD dataset. We visualize the classification map and semantic results in Fig. 18. After zooming in on Figs. 18(a) and 18(b), we can see that there are many details in the classification results, such as park trails, greenery, and building boundaries, which are all classified correctly. It can be seen that the UHR imagery has the ability to identify urban details. At the same time, the complex urban scenes have a lot of background noise, and individual objects (such as bridges and roads) have similar spatial and spectral characteristics, resulting in poor classification effects. We selected nine sub-regions in the test set to demonstrate the details of HRNet-W18 in Fig. 19, which is compared with ChangeMask. HRNet-W18 extracts the boundaries of the changed areas more accurately, and effectively distinguishes the changed objects, making the results of the classification and change detection better than those of ChangeMask. HRNet-W18 has a greater discriminative ability to deal with UHR images with pseudo-changes caused by the complex background. For example, the fourth and fifth lines of Fig. 19 have a lot of changed objects that are not of interest (e.g., cars and solar panels on the roofs), which are successfully recognized as no-change by HRNet-W18. The last two lines of Fig. 19 show the buildings demolished in 2018 and reconstructed with different structures in 2019, which is a typical change of buildings in an urban area. However, these two models do not recognize this change. There are also some obvious classification errors in Fig. 19, where bare land is confused with construction site, which is in the class of "other". Overall, HRNet can fuse the multi-resolution features to learn high-resolution representations of ground objects, resulting in a better performance.

6. Conclusion

In a complex and diverse urban area, SCD through the use of remote sensing images is a very meaningful and challenging task. Large-scale benchmark datasets have the potential to be combined with advanced technologies (such as deep learning) to promote the development of SCD. In this article, we have introduced a new large-scale UHR aerial imagery change detection dataset named the Hi-UCD dataset, which has rich semantic annotations, to detect more details of urban change. The Hi-UCD dataset can meet the requirements of UCD and its application with deep learning methods, because it has the following three characteristics:

- Large-scale data volume. The large amount of data provides a wide range of feature sources for deep learning methods, to improve the generalization.
- Semantic annotation. The rich semantic annotation allows the Hi-UCD dataset to be used to carry out multiple tasks at the same time, such as classification, BCD, SCD, etc.
- Ultra high resolution. The ultra-high spatial resolution truly restores the real world, while revealing more urban details, and provides the possibility for refined change detection.

Based on the Hi-UCD dataset, we selected 11 public deep learning based methods, including six BCD methods and five SCD methods, to establish a benchmark for UCD in binary and semantic change detection tasks, which proved that the Hi-UCD dataset is a comprehensive and



(a) classification map for 2018

(b) classification map for 2019

(c) semantic change map for 2018

(d) semantic change map for 2019

Fig. 18. The result of HRNet-W18 on the entire test region, which covers 54 km^2 with 540 million pixels. Legend: Water, Grass, Building, Greenhouse, Road, Bridge, Bare land, Woodland, Other, no-change. This figure is best viewed digitally with zoom. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

challenging dataset. The semantic consistency between changed and unchanged areas is very important in the change detection task. The metrics we proposed can be used to evaluate the performance of each method with regard to semantic consistency, from binary to semantic change. We also investigated the main challenges of the Hi-UCD dataset through experiments:

- Although there is serious sample imbalance in the Hi-UCD dataset, the unchanged samples play an important role in change detection. The unchanged pixels can help the models to identify the changed areas, and the semantic labels, if available, can also improve the classification accuracy.

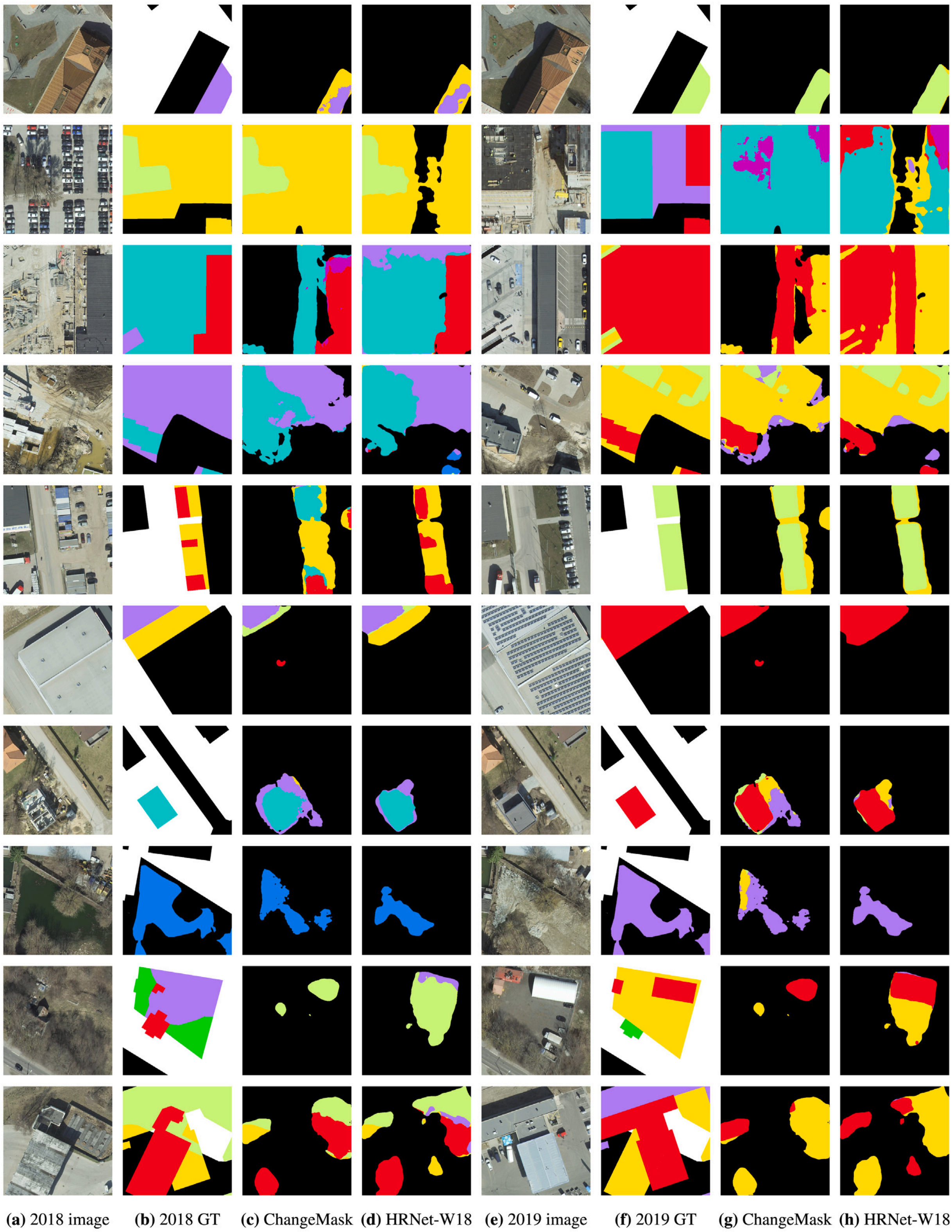


Fig. 19. Visualization of the sub-regions to demonstrate the details of HRNet-18, which is compared with HRSCD Str.4. Legend: Water, Grass, Building, Greenhouse, Road, Bridge, Bare land, Woodland, Other, no-change, and white is unlabeled. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

- HRNet performs better than other similar SCD methods. Improving the model's ability to extract features from high-resolution images can effectively improve the detection effect.

From the results of the experiments, we believe that how to use the UHR features to improve the semantic consistency in multi-task structure based SCD methods should be further considered. The severe sample imbalance in the semantic change classes also needs to be considered, because it is important for SCD.

In our future work, we will explore more effective deep learning methods to investigate the relationship between classification and change detection, with the aim being to obtain better semantic consistency. Furthermore, it is well-known that remote sensing data tend to suffer from various degradations, noise effects, or variabilities in the process of imaging. In addition, the spectral variability can cause non-interesting changes, which have an influence on the change detection. We could therefore incorporate the method proposed in Hong et al. (2018) to reduce the influence of non-interesting changes. The comprehensive experimental results will help researchers to make better use of the Hi-UCD dataset. We hope that the release of the Hi-UCD dataset will promote the development of UCD. More information abuot the dataset can be find in http://rsidea.whu.edu.cn/resource_sharing.htm or https://github.com/Daisy-7/Hi-UCD-S.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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